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On class imbalance in machine learning-based taxa identification: A comparative analysis of mitigation strategies

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ABSTRACT

Machine learning has transformative potential for automated monitoring of populations of insects and other arthropods, providing an important tool in biodiversity conservation. However, a major challenge persists: underrepresented species in training data lead models to favor common taxa and undermine efforts to track rare taxa that are important for various ecological studies. Although class imbalance mitigation strategies exist, their effectiveness in real-world image-based ecological contexts, particularly for rare species, remains understudied. In this study, a comprehensive comparison of widely used data-level, algorithm-level, and hybrid class imbalance methods was evaluated on two publicly available large-scale insect and invertebrate image datasets (FIN-Benthic2 and BIOSCAN). We performed an extensive empirical study of common techniques, including loss function adaptations, sampling strategies, and ensemble frameworks, using metrics tailored to imbalanced classification.

The key results demonstrate that Class Imbalance loss consistently outperforms alternatives, balancing rare and common species identification. Data augmentation and sampling methods exhibit dataset-specific efficacy, with no universal solutions. Hybrid approaches, particularly ensemble learning, yielded the strongest improvements, underscoring the value of integrating multiple strategies tailored to taxonomic granularity. Image generation using recent generative models was not able to improve classification performance in our experiments despite producing visually plausible images. Our findings emphasize context-dependent optimization, while providing actionable guidelines for deploying robust automated systems in biomonitoring programs. The code is available to support reproducibility and implementation. This study provides practical solutions to a central challenge in ecological machine learning, enabling more reliable tracking of rare species.

1. Introduction

Biomonitoring uses living organisms to assess environmental health, providing critical insights into ecosystem functioning and pollution impacts (Kang et al., 2024; Resh & Rosenberg, 1993). Human activities are causing rapid environmental degradation, resulting in species loss and ecological imbalances (Sagova-Mareckova et al., 2021). Insects and other arthropods serve as effective bioindicators for monitoring these changes, with freshwater macroinvertebrates, such as mayflies and caddisflies, exemplifying this role. These organisms possess characteristics ideal for biomonitoring: relatively long lifespans, limited mobility, and high sensitivity to habitat conditions. Their presence, absence, or changes in community composition can indicate pollution levels and other environmental disturbances. These patterns reveal long-term ecosystem stability, establishing macroinvertebrates as critical tools for assessing freshwater ecosystem health. Zaghloul et al. (2020)

Traditional biomonitoring relies on manual morphological identification, where taxonomists classify species by physical traits through microscopic analysis, creating significant operational constraints. This method demands extensive expertise and resources, making it time-consuming, costly, and labor-intensive (Simović et al., 2024). The global decline in taxonomic expertise, caused by the training requirements, further limits scalability (Cordier et al., 2019). These challenges have driven the development of automated solutions, particularly image-based species identification powered by machine learning (Keck et al., 2017). Advances in deep learning with CNNs now enable high-precision species identification by detecting subtle morphological differences and learning complex patterns from labeled image data (Cruz et al., 2022; Kwon et al., 2024). This automated approach has emerged as a useful tool in ecological research and biomonitoring, offering cost-efficient alternatives particularly for challenging sample

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types, such as bulk invertebrate collections, where manual sorting represents substantial workloads (Høye et al., 2022).

Machine learning models show promise for species classification, but a persistent challenge in ecological datasets is *class imbalance*, where samples are unequally distributed among classes (S. Chen et al., 2024; Schneider et al., 2020). This imbalance is not merely an artifact of data collection but reflects natural biodiversity patterns, where ecosystems typically comprise few common species and many rare ones, resulting in "long-tail" distributions (W. Chen et al., 2024). Due to these natural differences in species abundances, ecological image datasets are inherently highly imbalanced (Mohammadisabet et al., 2025; Xu et al., 2022). The number of training samples per taxon can vary from tens of thousands to just a few individuals (Raitoharju, Riabchenko, et al., 2018; Sohrab & Raitoharju, 2020). This skewed distribution poses significant challenges for machine learning classifiers, which can develop a bias toward the prevalent majority classes during training, resulting in poor predictive performance for underrepresented minority classes (Villon et al., 2022).

This bias is particularly problematic as rare species are often of ecological or conservation significance, especially for rare, endangered, and newly emerging taxa, for which reliable automated identification is most urgently needed. In automated species identification systems, the risk of misclassification is highest for rare species, those with the lowest abundance in a community, and can lead to false absences or underestimation of species richness, directly affecting ecosystem health assessment and conservation decisions (Villon et al., 2020). Recent work has also shown that AI systems trained on imbalanced biodiversity data exhibit systematic taxonomic biases that under-represent endangered and uncommon species (de Pedro Noriega et al., 2025), reflecting the same mechanism studied here. A second critical management domain is the early detection of invasive species: at the time of first arrival, an invasive taxon is by definition rare in the local community, and a classifier biased toward abundant native taxa will systematically suppress it, delaying the management response at the most cost-effective intervention window (Martinez et al., 2020). Reliable detection of rare species under conditions of limited and imbalanced training data is therefore not only a machine learning challenge but a direct prerequisite for timely conservation and biosecurity action (Belmont et al., 2022; Martinez et al., 2020).

In such severe class imbalance settings, overfitting can also become a serious issue because the model sees only a limited number of samples from the rare classes and, therefore, is more susceptible to memorizing irrelevant details instead of finding the real discriminating features. As a result, the generalization ability of the model is reduced, particularly for minority classes (Francazi et al., 2023). Furthermore, classes tend to be very fine-grained while containing considerable intra-class variance (Venkataramanan et al., 2021). The challenges intensify when models must simultaneously preserve subtle morphological features crucial for taxonomic identification while learning to accommodate natural variations within species.

Despite the importance of the problem, many ecological studies have avoided class imbalance by excluding rare species and using only balanced datasets with roughly equal sample sizes (Høye et al., 2022; Milo˘ sević et al., 2020; Sidumo et al., 2022). This artificial balancing undermines the applicability of models to real-world biomonitoring, where rare species are primary targets for conservation assessment and environmental change detection. Compounding this limitation, studies including imbalanced data often rely on overall accuracy as the primary metric (Ärje et al., 2020; Raitoharju & Meissner, 2019), which is misleading for imbalanced datasets (Farhadpour et al., 2024) as accuracy can be artificially inflated by predicting the majority class, masking inability to detect rare, ecologically critical species. Furthermore, there is no comprehensive comparative analysis evaluating diverse machine learning imbalance mitigation strategies for fine-grained ecological image-based classification with severe imbalance. This gap highlights the critical need for studies that apply diverse imbalance mitigation techniques in ecological contexts and evaluate their effectiveness using metrics designed specifically for imbalanced classification.

This study conducts a comparative empirical analysis of diverse class imbalance mitigation strategies for fine-grained ecological image-based taxa identification, particularly for insects and other arthropods. We evaluate commonly used data-level (upsampling, downsampling, augmentation, Generative Adversarial Networks (GANs)), algorithm-level (loss functions), and hybrid (ensemble) strategies, deliberately mainly focusing on widely-used baseline techniques rather than recent variants. Therefore, the conclusions of this study should be interpreted in the context of the evaluated baseline imbalance mitigation methods, particularly GAN-based generative augmentation, and should not be generalized to more recent methods or to diffusion-based generative models that were not tested here. By evaluating these established methods on two distinct ecological image datasets with severe imbalance, using metrics designed for imbalanced classification, our work provides empirical insights into their relative strengths and limitations in this specific domain. To the best of our knowledge, a comparative analysis of this kind has not been previously presented for fine-grained ecological image-based classification under such severe imbalance. By providing actionable guidelines based on comprehensive evaluation, this research contributes empirical evidence toward developing robust Artificial Intelligence (AI)-driven ecological monitoring systems, enhancing conservation efforts through more reliable identification of vulnerable and rare species. The study thus bridges the gap between general imbalance mitigation methods and their practical application in image-based biodiversity monitoring context.

The subsequent sections of this paper are structured as follows: Section 2 reviews related work. Section 3 details the underlying machine learning methodologies and the specific class imbalance strategies investigated in this study. Section 4 outlines the experimental design, describes the datasets, and presents the results of the comparative analysis. Finally, Section 5 discusses the implications of our findings and proposes avenues for future investigation.

2. Related work

In this section, we review related work on data imbalance in image-based ecological studies and general comparisons of imbalance mitigation methods. Note that here we only give a high-level overview of imbalance mitigation methods that have been applied in some ecological classification task. The details of the methods are discussed in Section 3.

2.1. Class imbalance in fine-grained ecological image-based taxa identification

Although most of the fine-grained ecological image-based taxa identification studies ignore the class imbalance problem, there are some works that have acknowledged and tried to address the problem. A systematic review of automated insect detection systems by Teixeira et al. (2023) revealed a critical methodological gap in insect detection and classification. Their analysis of 36 studies identified class imbalance as the predominant factor influencing detection performance. Despite recognizing synthetic data generation and Focal loss as potential mitigation strategies, the authors acknowledged that these methods lack thorough empirical validation in the context of insect detection. It is important to note that this analysis was based on a systematic review only and did not conduct a direct comparison.

Miller et al. (2025) conducted a comprehensive systematic review of AI applications in aquatic biodiversity research, identifying multiple methodological challenges, including issues related to data quality and bias. Among their findings, they noted that training datasets often overrepresent common species while underrepresenting rare and endangered taxa, potentially compromising model performance for conservation-critical species. Their review encompassed various deep

learning architectures, including CNNs (ResNet, YOLO, EfficientNet) for image-based species identification, highlighting the need for improved data diversity and standardized methodologies. While discussing multiple technical challenges, the authors emphasized transfer learning and data augmentation as promising approaches for enhancing model generalization across diverse aquatic ecosystems and addressing data imbalance issues affecting rare and endangered species.

Mohammadisabet et al. (2025) implemented a comprehensive dual-strategy approach to address severe class imbalance inherent in fish species datasets. Their methodology integrates sophisticated data augmentation techniques with synthetic sample generation and style transfer methods, effectively expanding the dataset from 3960 to 7290 images while reducing the imbalance ratio from 30:1 to 6:1. This systematic intervention yielded an improvement in recall for underrepresented species and elevated overall model accuracy.

Xu et al. (2022) developed a peer learning network specifically engineered to mitigate class imbalance in fine-grained butterfly recognition. Their methodology addresses the difficulty of learning robust and discriminative features while mitigating bias and variance problems characteristic of long-tailed classifiers. The architecture employs a distribution-aware penalty mechanism that selectively utilizes training samples based on distribution characteristics through a knowledge exchange strategy. This distribution-aware approach achieved significant performance improvements across all class categories, demonstrating the efficacy of specialized architectural implementations for addressing long-tailed distributions in ecological datasets.

Binta Islam et al. (2023) tackled class imbalance in herpetofaunal species classification using camera trap images. They employed systematic oversampling through geometric transformations, applying 180-degree and 90-degree rotations to minority class images (snakes and toads) to normalize the class distribution to 1250 samples per category. This approach, combined with additional data augmentation techniques (zooming, rotation, shearing, and horizontal shifting), enabled robust classification despite challenges, such as small body sizes and natural camouflage. Their results showed that pretrained architectures with fine-tuning, when applied to the balanced dataset, achieved superior performance in terms of accuracy and F1-score, establishing benchmarks for automated herpetofaunal monitoring systems.

Simões et al. (2023) used data augmentation to address class imbalance in wildlife monitoring tasks. They created new training data by horizontally flipping existing images. This helped the model learn more robust features and reduced overfitting, improving performance despite the imbalanced dataset.

Simović et al. (2025) examined class imbalance effects on CNN-based identification of aquatic insects using EfficientNetB2. They tested scenarios from balanced to severely imbalanced datasets, mirroring natural species distributions. Their analysis revealed accuracy degradation due to class imbalance, particularly affecting minority species. Random oversampling significantly improved performance in imbalanced models. The study quantified imbalance impacts on deep learning in ecology, validated oversampling as a mitigation strategy, and suggested a minimum of 30 samples per class for robust classification.

In a study on automated insect detection (Bjerge et al., 2023), Focal loss was applied to address the substantial class imbalance inherent in ecological datasets, where species representation varied from 286 to 6934 training samples across nine taxonomic categories. This adaptive loss formulation enabled the YOLOv5 model to achieve satisfactory average precision by dynamically reducing the contribution of well-represented species like *Apis mellifera* while emphasizing learning from rare species such as *Eristalis tenax*. However, there were no comparisons between Focal loss and other classification losses. Focal loss was applied yielding improvement compared to Cross-entropy loss for minority species identification, while acknowledging a trade-off with a slight decrease in accuracy for majority species (Goldsworthy & Matam, 2021), making Focal loss as a viable solution for addressing conservation-critical challenges in automated species monitoring systems where rare taxa detection supersedes aggregate accuracy metrics.

Table 1 summarizes studies that identify and discuss class imbalance as a challenge in the context of CNN-based ecological classification. The table demonstrates the widespread recognition of this problem within the relevant scientific literature.

2.2. General comparisons of class imbalance mitigation strategies

General-purpose studies examining class imbalance methods in CNN-based models offer a comparative framework for evaluating their applicability to ecological taxa identification. Contemporary perspectives are provided by a comprehensive survey, which categorizes mitigation strategies into data-level approaches (upsampling, downsampling, augmentation), algorithm-level approaches (cost-sensitive learning, Focal loss, etc.), and hybrid methods (Saini & Susan, 2023). This taxonomy underscores both the continued relevance of established techniques and emerging deep learning innovations. The same categorization is also followed in our study.

The foundational study by Buda et al. (2018) systematically examined imbalance effects on CNN using upsampling, downsampling, two-phase training, and thresholding on MNIST, CIFAR-10, and ImageNet datasets. Their findings demonstrated that classification performance deteriorates with increasing imbalance severity and dataset complexity, with upsampling consistently proving most effective while downsampling remained effective but computationally efficient for extreme imbalance scenarios, where minority classes often have several orders of magnitude fewer samples than majority classes. A recent survey on resampling strategies found that their effectiveness depends strongly on the application requirements, and that no single method provides superior performance across all applications (Carvalho et al., 2025). The authors suggest detecting and quantifying existing data difficulty factors, including class overlap, small disjuncts, and noise, that bias the models toward the majority class and then focusing the sampling efforts on these problematic data regions.

In medical imaging applications, a study evaluated standard methods on high-resolution mammography images, finding that while imbalance mitigation reduced majority class bias, overall predictive performance improvements were inconsistent (Walsh & Tardy, 2023). Their proposed synthetic lesion generation approach demonstrated superior generalization capabilities, suggesting potential for hybrid strategies (combining synthetic lesion generation with traditional class imbalance techniques, such as over-sampling or class weighting) in severe imbalance scenarios.

An empirical evaluation presents a critical counterpoint (Kim & Hwang, 2022). Assessing sampling methods across 31 datasets revealed that sampling frequently degraded classifier performance, with undersampling showing particularly poor results. This challenges the assumption that re-balancing inherently improves outcomes. Yang et al. (2022) conducted an ecologically important study of the class imbalance problem while identifying empty camera trap images. They suggested different training set construction approaches for different class imbalance ratios. However, their study did not consider imbalance in taxa identification, only the binary imbalance between empty and non-empty camera trap images.

Recent surveys highlight GANs as sophisticated data-level approaches capable of generating realistic synthetic images for minority class augmentation, potentially addressing limitations of simpler methods, such as Synthetic Minority Oversampling Technique (SMOTE) (Hung & Gan, 2021; Sampath et al., 2021). Nevertheless, GAN training instability and inconsistent effectiveness remain active research challenges (Nazari & Branco, 2021). The survey by Sampath et al. (2021) points out that when using GANs in image generation for problems with class imbalance, users must be cautious about the quality of generated images, as low-quality samples may introduce noise into the dataset. Furthermore, we did not find any GAN-focused survey that would combine severe imbalance and fine-grained classes, which are the focus of our study.

Table 1
Overview of class imbalance issues in CNN-based ecological classification literature.

Source	Ecological context	Imbalance ratio	Imbalance mitigation
Teixeira et al. (2023)	Insects (general, agricultural pests from traps)	—	Data augmentation, Focal loss (recommended)
Miller et al. (2025)	Aquatic species (general; freshwater)	—	Data augmentation, transfer learning; call for diverse and annotated datasets
Mohammadisabet et al. (2025)	Fish species (marine)	1 : 30	Data augmentation, Transfer learning
Xu et al. (2022)	Insects (Butterflies, fine-grained)	—	Specialized architecture and loss, distribution-aware penalty mechanism
Binta Islam et al. (2023)	Herpetofauna (toads/frogs, lizards, snakes from camera traps)	~1 : 1 : 3	Oversampling (image rotation), Data augmentation
Simões et al. (2023)	Wildlife Dataset	–	Data augmentation
Simović et al. (2025)	Aquatic insects (Ephemeroptera, Plecoptera, and Trichoptera)	1 : 8	Oversampling technique significantly improved accuracy
Bjerge et al. (2023)	Insect Species	~ 1 : 24	Focal loss
Goldsworthy and Matam (2021)	Bird Species	1 : 124	Focal loss

W. Chen et al. (2024) conducted a recent survey on imbalanced learning approaches and list several papers, where different combinations of imbalance mitigation approaches have worked well. They also suggest future research topics for imbalanced scenarios, including adaptive and active learning approaches. Beyond resampling and loss-based strategies, a substantial body of work has advanced long-tail recognition through decoupled training frameworks (Zhou et al., 2020), feature-level synthesis for tail classes (Wang et al., 2021), and prototype-based contrastive learning (Cui et al., 2021); a comprehensive survey of these directions is provided by Zhang et al. (2023). These methods fall outside the scope of the present study, which deliberately focuses on widely used baseline imbalance mitigation strategies applicable across ecological monitoring pipelines.

3. Methods

This section outlines the machine learning methodologies, class imbalance mitigation strategies, and evaluation metrics investigated in our comparative study. We first briefly discuss the most relevant machine learning approaches, including CNNs, transfer learning, and data augmentation. We then describe specialized techniques to address data imbalance and conclude by detailing the evaluation metrics used to assess model performance.

3.1. Underlying machine learning approaches

3.1.1. Convolutional neural networks

Convolutional Neural Networks (CNNs) are a special kind of deep learning system created to work with data that is arranged in a grid, like the pixels in an image. The ability of CNNs to learn hierarchical features automatically has revolutionized the computer vision field in general (Raitoharju, 2022) and is expected to revolutionize various ecological studies (Høye et al., 2021). CNNs typically consist of three types of layers: convolutional layers, pooling layers, and fully connected layers (Yang & Li, 2017). Convolutional layers are responsible for learning important features from images. Different convolutional layers learn different features during training (Ghosh et al., 2020; Yu et al., 2016). For example, lower layers learn low-level features, such as edges, simple textures or corners. The middle layers combine learned features from lower layers and detect more complex features, such as complete objects or some parts of it. The final layers extract more abstract features and capture complete objects or full scenes in images by using already learned complex features. Pooling layers, also known as downsamplers, are introduced to reduce spatial dimension, which helps reduce overfitting. Fully connected layers are used for classification or regression tasks. Recent advances in CNN architectures have further improved the success of CNNs in various fields, such as image classification, object detection, image segmentation, and face recognition (Li et al., 2022). While CNNs' ability to learn to extract meaningful features directly from the data is their main advantage, this also requires a lot of training data. This is a factor contributing to the class imbalance problem: it is challenging to learn a good representation from limited training data. One approach to tackle this challenge is transfer learning.

3.1.2. Transfer learning

Transfer learning is a powerful method in which a model is first trained using a large amount of available data for a specific task (Zhao et al., 2024). Then, it is fine-tuned for a related task with a lower learning rate. Transfer learning has been successfully utilized in many applications, especially for tasks relevant to a similar domain, but with limited labeled data. It uses the information learned from a large dataset and, in turn, avoids extensive retraining of the model for the task at hand, which has limited labeled data. The success of early CNN architectures, such as AlexNet, in large-scale challenges demonstrated the potential of transfer learning, marking a turning point in deep learning for computer vision (Krizhevsky et al., 2012). Subsequently, various other architectures have been proposed and utilized for transfer learning, and its effectiveness has been demonstrated in numerous studies, particularly in applications with limited data availability when compared to traditional machine learning methods (Valan et al., 2019; Zhuang et al., 2021).

3.1.3. Data augmentation

Data augmentation is commonly used with image data. It leverages various transformations, such as flipping, cropping, rotation, and contrast adjustment, among others, to increase the number of training samples using existing images (Arun Pandian et al., 2019; Wang & Perez, 2017). Data augmentation, thus, increases the diversity of images without the need for collecting new images. Data augmentation is commonly used also in species identification to improve classification performance. In a study on acoustic species identification of 18 small mammal species, MacIsaac et al. (2024) demonstrated that data augmentation can significantly improve model performance, especially when original training samples are scarce, highlighting its potential for developing classifiers for species where data is limited or difficult to collect. Numerous studies have shown that artificially increasing data using augmentation can improve performance (dos Santos & Gonçalves, 2019; Sun et al., 2022). To address the challenges of class imbalance in their dataset, Simões et al. (2023) employed data augmentation. This was achieved by applying a horizontal flip to randomly selected images, thereby generating new training samples. The authors noted that this

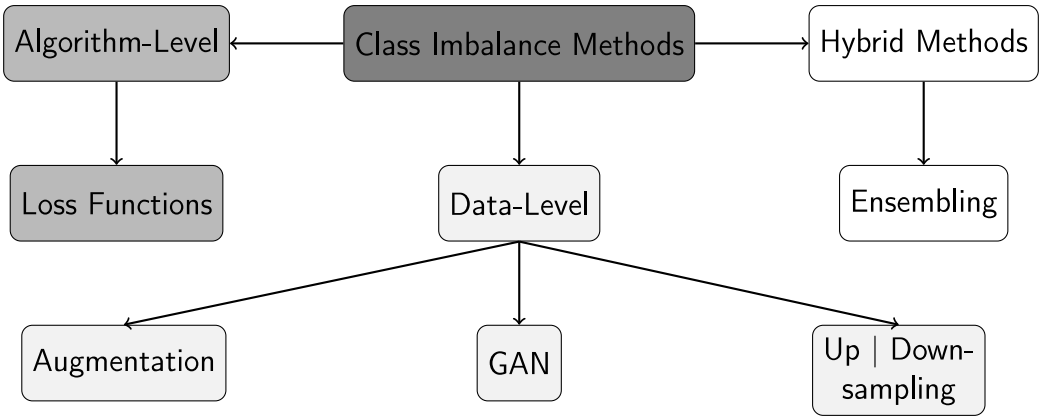


Fig. 1. Hierarchical taxonomy of class imbalance mitigation strategies.

technique was used to increase the model's ability to generalize with more robust features and to reduce overfitting, which are common issues when dealing with imbalanced data in ecological monitoring applications.

3.2. Class imbalance mitigation methods

Class imbalance mitigation methods are commonly categorized into three main groups: algorithm-level methods, data-level methods, and hybrid methods (Buda et al., 2018; Saini & Susan, 2023). Fig. 1 illustrates the three main approaches to handling class imbalance. Algorithm-level methods account for the class imbalance distribution by affecting the training process or decision strategy of the model. An example of algorithm-level method is cost-sensitive learning that assigns higher costs to minority classes during the training process. Data-level methods encompass techniques that directly manipulate data distribution to achieve balanced distribution. Examples include upsampling and downsampling. Hybrid methods combine data and algorithm-level approaches to enhance performance.

3.2.1. Algorithm-level methods

Algorithm-level methods modify the learning algorithm itself or its learning process to make it more sensitive to minority classes, without altering the original data distribution (Saini & Susan, 2023). This is often achieved by adjusting the algorithm's objective function or decision-making process to account for the unequal class distribution or differing misclassification costs.

Cost-sensitive learning (CSL) acknowledges that misclassifying samples from different classes may incur different costs or consequences (Rahimi Mashkale et al., 2025). Instead of aiming to minimize the overall number of errors, CSL aims at minimizing the total misclassification cost (Ling & Sheng, 2008). In the context of class imbalance, this typically involves assigning a higher misclassification cost to errors made on the minority class samples (Araf et al., 2024). This forces the learning algorithm to pay more attention to correctly classifying these rare but important instances. **Loss function** design is a primary way to implement CSL or otherwise modify algorithm behavior for imbalanced data, as it guides the model's optimization process during training.

The standard *CE* loss, a widely utilized loss function in classification tasks, quantifies the divergence between predicted and actual probability distributions as follows:

$$\mathcal{L}_{\text{CE}} = -\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\,\ln\left(p_{i,c}\right). \tag{1}$$

Here, N denotes the number of samples in the training batch, C represents the number of distinct classes, $p_{i,c}$ is the predicted probability that sample i belongs to class c, and $y_{i,c} \in \{0,1\}$ is the one-hot ground-truth indicator (equal to 1 if sample i belongs to class c, and 0 otherwise). Application of CE loss on imbalanced datasets presents significant challenges, potentially inducing model bias toward majority classes. This inherent limitation stems from its uniform treatment of samples, where equal weight is assigned to all instances regardless of the underlying class distribution. To address this methodological constraint, researchers have proposed numerous alternative loss functions specifically designed for scenarios characterized by imbalanced class distributions.

Focal loss handles class imbalance challenges by introducing an adaptive weighting mechanism that dynamically adjusts the contribution of each sample based on its classification difficulty (Lin et al., 2017). This formulation extends the conventional CE loss by incorporating a modulating factor that automatically down-weights the contribution of well-classified samples while emphasizing the challenging instances during training. The mathematical representation of Focal loss can be expressed as follows:

$$\mathcal{L}_{\text{Focal}} = -\sum_{i=1}^{N}\sum_{c=1}^{C} \alpha_{i,c}(1-p_{i,c})^{\gamma}\, y_{i,c}\,\ln(p_{i,c}), \tag{2}$$

where $\alpha_{i,c}$ is a balancing factor for class weighting, $(1-p_{i,j})^{\gamma}$ serves as the modulating factor, γ is the focusing parameter that controls the rate of down-weighting, $p_{i,c}$ denotes the predicted probability for class c of sample i, and $y_{i,c}$ indicates the ground truth label. This formulation enables the model to focus more on difficult minority class samples while naturally reducing the overwhelming influence of easily classified majority class instances. The focusing parameter γ provides flexibility in controlling the extent of this modulation, allowing for task-specific optimization of the learning process.

While Focal loss has given promising results in some ecological applications (Bjerge et al., 2023; Goldsworthy & Matam, 2021) and shows potential to address issues caused by class imbalance, it faces inherent limitations when handling datasets with extreme class imbalances. Focal loss is limited when handling significant class imbalance as it fails to consider large differences in sample counts among classes. Specifically, it uses the same focusing parameter, for all categories, which means it does not adequately address the imbalance between positive and negative samples for rare categories (Islam et al., 2024; Yang et al., 2025). This can cause the model to over-train on frequent yet difficult samples, leading it to ignore rare ones. While Focal loss prioritizes hard-to-train samples, it may not adequately accommodate rare-class samples, resulting in suboptimal performance on imbalanced datasets (Su et al., 2025).

Class Imbalance (CI) loss, proposed by Yue et al. (2023), introduces a dynamic weighting mechanism that considers both class frequency and prediction confidence. The mathematical formulation of CI loss is defined as follows:

$$\mathcal{L}_{\text{CI}} = -\sum_{i=1}^{N}\sum_{c=1}^{C} \left(\log\frac{N_{\max}}{N_c} + \theta\right) e^{-k p_{i,c}}\, y_{i,c}\,\log(p_{i,c}), \tag{3}$$

where $N_{\max}$ denotes the majority class sample count, N_c denotes the number of samples in class c, and hyperparameters k and θ control loss adjustment and weight normalization, respectively. The original authors of the method report that, based on their empirical validations, optimal performance is demonstrated with hyperparameters set to $k = 1$ and $\theta = 0.5$. CI loss function establishes an inverse relationship between class frequency and weight allocation — majority classes receive lower weights while minority classes gain higher importance in the optimization process. This weighting scheme enhances minority class contribution during gradient computation, effectively addressing the inherent bias toward majority classes. The logarithmic term $\log(N_{\max}/N_c)$ maintains reasonable weight differences across classes, while the addition of θ ensures non-zero weighting factors.

Empirical evaluations on medical imaging benchmarks have demonstrated the superiority of CI loss over established alternatives. For instance, in endoscopic image classification tasks (Yue et al., 2023), CI loss consistently outperformed CE loss, Focal loss, and Dynamically Weighted Balanced Loss across multiple evaluation metrics. On a polyp recognition dataset with class imbalance (18,643 normal vs. 4292 polyp images), CI loss yielded substantial improvements in minority class identification while maintaining overall system performance. The method showed particular efficacy in handling hard samples through its exponential weighting term, $e^{-kp_{i,c}}$, which adaptively adjusts sample importance based on prediction confidence. Furthermore, comparative analysis on the multi-class Hyper-Kvasir dataset (10,662 images across 23 categories) revealed robust performance for CI loss in extreme imbalance scenarios, achieving superior performance compared to contemporary loss functions. This performance advantage stems from the loss function's dual consideration of class frequency and prediction difficulty, enabling effective optimization across heterogeneous class distributions without gradient explosion issues commonly observed in alternative formulations. To our knowledge, no prior studies have applied CI loss to ecological taxa identification.

3.2.2. Data-level methods

Data-level strategies for CNN involve various methods designed to balance the class distribution by adjusting the number of majority and minority class samples prior to model training (Duarte et al., 2020). The primary objective is to present the learning algorithm with a more balanced representation of the classes, thereby reducing the inherent bias toward the majority groups. Sampling techniques are among the most common data-level approaches. They directly alter the dataset composition by either increasing the number of minority class instances or decreasing the number of majority class instances.

Downsampling technique involves removing samples from the majority class to match the number of minority class samples or to achieve a target ratio, then using CNN for classification (Lee & Bang, 2022). Downsampling has also seen application in ecological research, but only with traditional machine learning approaches (Benkendorf et al., 2023; Chen et al., 2004). In these studies, downsampling improved the classification performance, but the results may be different for deep learning -based approaches that generally require more training data than the traditional approaches. Some non-ecological studies suggest that downsampling can help reduce the risk of overfitting associated with large majority classes (Lee & Bang, 2022).

Upsampling method duplicates minority class images or creates new samples for the minority class using data augmentation techniques (See Section 3.1.3) until a desired balance ratio is achieved, which when combined with CNNs has demonstrated enhanced classification performance on imbalanced datasets (Abdani et al., 2022; Joloudari et al., 2023). Upsampling has been widely adopted in various studies. For instance, Wahab et al. (2017) used upsampling and ensemble methods combined with a two-phase CNN approach to handle class imbalance in histopathological breast cancer detection. Other studies also confirm the general utility of upsampling for improving model performance on imbalanced data (Huang & Dai, 2021). For detecting marine litter from Unmanned Aerial System (UAS) images, upsampling was preferred over downsampling (Duarte et al., 2020). This was done to avoid losing important data, and it resulted in superior performance. While upsampling can improve performance, the research on camera trap image classification demonstrates that upsampling may not always be helpful. The study found that neither upsampling nor downsampling reduced model errors compared to the baseline non-sampling method for complex datasets, indicating that the effectiveness of upsampling depends on dataset characteristics and complexity (Yang et al., 2021).

In agricultural applications, which often overlap with ecological monitoring, data augmentation techniques (including generative methods) have been employed for tasks, such as crop classification and object detection, often specifically to alleviate data imbalance and improve performance in remote sensing applications (Sharma & Gosain, 2025). While standard geometric and color augmentations are common preprocessing steps in many image-based ecological studies, their specific role in addressing class imbalance, beyond simply increasing the overall dataset size, is often achieved by employing them as a mechanism for oversampling the minority class (Duarte et al., 2020) or by using more advanced generative techniques, such as GANs designed to synthesize minority samples (Rahman et al., 2024).

Generative Adversarial Networks (GANs) (Goodfellow et al., 2014), a recent advancement in generative artificial intelligence, offer powerful models for creating synthetic data. These models present a promising alternative to traditional oversampling methods for image data, as they can generate novel samples to help balance the dataset. A GAN consists of two networks: a generator (G) and a discriminator (D), which are trained in an adversarial process. The generator creates synthetic samples $G(z)$ from random noise z, while the discriminator learns to distinguish them from real samples x.

The networks are optimized via the following objective function (Goodfellow et al., 2014):

$$\min_G \max_D V(D,G) = \mathbb{E}_{x\sim p_{data}(x)}[\log D(x)] + \mathbb{E}_{z\sim p_z(z)}[\log(1 - D(G(z)))], \quad (4)$$

where G aims to minimize the objective while D aims to maximize it and $p_{data}(x)$ is the distribution of real data and $p_z(z)$ is the input noise distribution.

GANs have been applied to various domains with class imbalance (Shamsolmoali et al., 2021). In agriculture, GANs aid in image augmentation for tasks such as plant and animal disease detection, plant classification, and wildlife detection using thermal imagery generated from color images (Rahman et al., 2024). The Deep Convolutional Generative Adversarial Network (DCGAN) (Radford et al., 2016) is designed to learn hierarchical representations from image data and generate new images in an unsupervised manner. Several studies highlight the use of DCGANs to address data or class imbalance in image datasets. For instance, DCGAN was utilized for data augmentation to generate synthetic minority class samples in a breast cancer dataset, achieving improved performance (Saini & Susan, 2020). Wu et al. (2020) employed DCGAN for data augmentation to identify tomato leaf diseases. They generated synthetic images to expand a small, manually collected dataset, thereby improving the recognition accuracy and generalization ability of a GoogLeNet-based classifier.

Arjovsky et al. (2017) introduced the Wasserstein Generative Adversarial Network (WGAN) to improve the training stability of GANs by using Earth-Mover (EM) distance to measure the difference between real and generated data distributions. Bhatia and Dahyot (2019) used WGAN and showed that it improved classifier performance under class imbalance. Karras et al. (2018) presented Progressive Growing GAN (ProGAN), where the generator and discriminator start with low-resolution images and progressively increase the spatial resolution by incrementally adding layers. Abdelhalim et al. (2021) applied a ProGAN-based method for generating fine-grained skin lesion images and showed improved detection performance. Karras et al. (2019) proposed StyleGAN, a style-based generator architecture in which the input noise is first mapped to an intermediate latent space and then

transformed into a style through a learned affine transformation (Wang et al., 2020). This architecture improved control over image synthesis and supported the generation of more realistic fine-grained details. StyleGAN2 (Karras et al., 2020) further improved the architecture by replacing adaptive instance normalization with weight demodulation and by using skip connections and perceptual path length regularization. Building on this, StyleGAN3 (Karras et al., 2021) addressed texture sticking and aliasing issues by redesigning the generator to better preserve consistency under image transformations (De Souza et al., 2023). Klasen et al. (2022) tested StyleGAN-based augmentation and observed improved identification performance for underrepresented species of scarab beetles, bees, and butterflies.

Despite their potential, GANs represent a relatively complex technique with several challenges, and their training can be complex and unstable (Huang et al., 2022). While GAN-based oversampling is recognized as a promising preprocessing method in some fields (Kupas & Harangi, 2021), its application in core ecological tasks has not been systematically validated. Generating high-quality minority samples for complex image-based ecological data remains a significant challenge. Further research demonstrating clear advantages over simpler, well-established methods in diverse ecological contexts is needed before they can be broadly adopted. More recently, diffusion-based generative models have emerged as an alternative for synthetic minority-class image generation, showing improved image fidelity and diversity compared to GANs (He et al., 2025). However, diffusion-based approaches are not evaluated in the present study; as noted in the introduction, conclusions regarding generative augmentation apply specifically to the GAN-based methods examined here.

3.2.3. Hybrid methods

Hybrid methods address the class imbalance by combining multiple fundamental approaches of data-level methods, algorithm-level methods, or both. This integrated approach maximizes the advantages of multiple strategies while minimizing their individual limitations (Galar et al., 2012). Among various hybrid methodologies, ensemble learning frameworks have emerged as a particularly effective and widely used approach for addressing class imbalance challenges (Buda et al., 2018). These frameworks combine predictions from multiple base models to improve classification robustness in imbalanced scenarios.

Ensemble learning is a key hybrid strategy that combines predictions from multiple base models to improve classification robustness and generalization in imbalanced scenarios by leveraging diverse model outputs to address skewed class distributions (Liu et al., 2017; Thabtah et al., 2020). A prominent example is *Bagging*, which utilizes bootstrapped sampling (random subsets with replacement) to train independent models on distinct data variations (Khan et al., 2024). The sampling with replacement means that some data points may appear multiple times in a single subset, while others may not be included at all. This process is crucial because it introduces diversity among the base models; because each model is trained on a slightly different dataset, they learn slightly different features and make different errors. By aggregating the predictions of these diverse models, Bagging reduces the variance of the final prediction, leading to a more stable and accurate classifier that is less prone to overfitting on the training data.

While various pooling functions exist for combining predictions (such as majority voting and weighted averaging), averaging is a common and effective approach where the final prediction is determined by averaging the probability scores from all ensemble members (Zhou, 2021). By integrating the confidence scores (i.e., probabilities) from all participating models, this method enhances the reliability of the final prediction, particularly in imbalanced classification scenarios where it can help moderate the influence of any single overconfident model. The final ensemble probability for class y given input x is calculated as follows:

$$P(y|x) = \frac{1}{M}\sum_{i=1}^{M} P_i(y|x), \tag{5}$$

where $P_i(y|x)$ is the probability predicted by the ith model among a total of M models in the ensemble.

3.3. Evaluation metrics

Most machine learning methods assume balanced class distributions, leading to biased models that prioritize common classes when applied to imbalanced data (Thabtah et al., 2020). This bias reduces accuracy for rare species, often the most ecologically critical classes, as models focus excessively on dominant patterns. The problem worsens when rare classes overlap with common ones in feature space or contain internal imbalances (e.g., intra-taxon variations) (Sohrab & Raitoharju, 2020). Standard training processes, such as gradient updates in neural networks, naturally favor majority classes due to their numerical dominance (Sun et al., 2009). Common metrics, such as accuracy, fail to capture these issues, as high scores can mask poor rare-class detection (Thölke et al., 2023). This underscores the need for evaluation metrics specifically designed to assess performance across all classes, particularly in ecological applications where rare taxa demand precise identification (de la Cruz Huayanay et al., 2025; Johnson & Khoshgoftaar, 2019; Thabtah et al., 2020).

In order to introduce classification metrics, we need the following key concepts:

TP = True Positives (correctly predicted instances of the positive class)

TN = True Negatives (correctly predicted instances of the negative class)

FP = False Positives (negative instances incorrectly predicted as positive)

FN = False Negatives (positive instances incorrectly predicted as negative)

$$\text{Precision} = \frac{TP}{TP+FP} \quad \text{(Positive Predictive Value)}$$

$$\text{Recall} = \frac{TP}{TP+FN} \quad \text{(True Positive Rate)}$$

The above definitions are based on a binary classification setup. In multi-class settings, these metrics are typically adapted using a one-vs-rest (OvR) approach, where each class is iteratively treated as the 'positive' class while all other classes collectively form the 'negative' class. After calculating the per-class metrics, the results are aggregated into a single score using one of several averaging strategies (Lim et al., 2024; Thölke et al., 2023):

- *Macro averaging* calculates the metric independently for each class and then takes the unweighted average. This gives equal weight to every class, regardless of its size, making it particularly useful for highlighting performance on rare classes.
- *Micro averaging* aggregates the contributions of all classes by summing their individual true positives, false positives, and false negatives before computing the metric. Because this approach treats every sample equally, it gives more influence to larger classes and is, thus, sensitive to class imbalance, often mirroring overall accuracy.
- *Weighted averaging* computes the metric for each class and then takes the average, weighted by the number of true instances for each class (the support). This provides a score that accounts for class imbalance but can still be dominated by majority classes.

Accuracy is a common metric for classification models, representing the proportion of total predictions that the model classified correctly (Owusu-Adjei et al., 2023). Its definition is

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}. \tag{6}$$

Accuracy can be misleading for imbalanced datasets because it is heavily influenced by the majority class (Tamouh et al., 2025; Thölke et al., 2023). This can lead to the "accuracy paradox", where a model achieves a high score by simply predicting the majority class, even if it fails to identify any minority class instances (van den Goorbergh et al., 2022; Wongvorachan et al., 2023). In ecological monitoring, this could mean a model appears successful while failing to detect rare or endangered species. Therefore, accuracy is not a reliable standalone metric for imbalanced problems, and metrics specifically designed for

imbalanced distributions are essential for a reliable assessment (Thölke et al., 2023).

F1-score balances precision (the ability to avoid false alarms) and recall (the ability to find all true cases) by calculating their harmonic mean as

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{7}$$

The use of the harmonic mean is crucial because, unlike a simple average, it is significantly more sensitive to low values, ensuring that a weakness in either precision or recall is not hidden (Jafarigol et al., 2023). This property makes the F1-score particularly well-suited for tasks with class imbalance. In a multi-class setting, the macro-averaged F1-score extends this benefit by first calculating the F1-score for each class independently and then computing their unweighted mean. This approach treats every class as equally important, regardless of its size. Consequently, a high macro-F1 score indicates that a model performs consistently well across all categories, successfully identifying instances of each class (high recall) without a high rate of misclassification (high precision). For a task like insect classification, the macro-F1 score ensures a model is rewarded only if it accurately detects both rare and common species, effectively penalizing models that achieve high performance on common species at the expense of failing to identify rare ones.

Balanced Accuracy (BACC) measures model performance by averaging recall values. The multi-class BACC is the macro-average of class-specific recalls. So in simple words, BACC focuses on the true positive rate of all classes, and if the method does not perform well on all classes, it is reflected in the BACC value.

$$\text{BACC} = \frac{1}{C} \sum_{c=1}^{C} \text{Recall}_c, \tag{8}$$

where C is the number of classes and Recall_i is the recall for cth class.

Geometric Mean (G-mean) evaluates classification balance by calculating geometric mean of recall score of all classes, heavily penalizing poor performance on any class, as a low score even in one recall value drastically lowers the overall value (Jafarigol et al., 2023). The G-mean is defined as

$$\text{G-mean} = \sqrt[C]{\prod_{i=1}^{C} \text{Recall}_c} \tag{9}$$

It ranges from 0 (worst) to 1 (best) and forces models to perform well on all classes. It is essential for ecological tasks where both rare species detection, such as endangered insects, and common species accuracy matter, preventing models from excelling on dominant classes while ignoring rare ones. Its advantages include class fairness, where high scores require good recall for every class, and imbalance resistance, being unaffected by skewed class sizes, with rare species having equal weight. Its limitations include zero collapse, where the score fails completely if any class has zero recall (Lemaître et al., 2017).

Matthews Correlation Coefficient (MCC) is a robust metric that summarizes the entire confusion matrix into a single value between −1 and +1. Unlike the F1-score or BACC, which are typically calculated by averaging binary results, the multiclass MCC treats the entire $K \times K$ confusion matrix as a single correlation problem (Gorodkin, 2004; Pedregosa et al., 2011). This metric is effective for imbalanced data because it only yields a high score if the model performs well across all categories, making it a reliable measure of a model's global predictive consistency.

Practical Guidance for Evaluation Metric Selection in Class Imbalanced Applications: For imbalanced ecological classification tasks, adopt macro-averaged BACC or F1-score to prioritize rare taxa detection. Always retain full confusion matrices during evaluation, as they enable recalculating metrics for specific classes or new criteria without reprocessing data or retraining models. Avoid micro-averaging, as it mirrors standard accuracy and disproportionately favors common species. Pair BACC and F1-score with G-mean to evaluate critical failures in individual classes. Furthermore, the MCC serves as a robust correlation measure for the entire confusion matrix, providing a reliable global indicator of predictive consistency that accounts for both common and rare classes more effectively than standard accuracy. Its inclusion ensures that improvements in minority-class detection do not come at the expense of sacrificing the model's overall structural reliability. These metrics are crucial as they enable evaluation of the model's performance comprehensively. For example, G-mean relies on recall of all classes, thus it is a critical metric to assess effectiveness in class imbalance problems. The F1-score balances precision and recall, a key property for analyzing predictive capability under class imbalance conditions. These metrics collectively offer a robust measure for model evaluation (Moorthy et al., 2024; Shanmugam et al., 2025).

4. Experiments and results

In this section, we present an empirical study evaluating class imbalance mitigation strategies on two ecological image datasets. Using a comparative framework, we assess data-level, algorithm-level, and hybrid approaches under conditions of significant class imbalance, similar to those found in real-world biodiversity monitoring scenarios.

4.1. Experimental framework

This section outlines the methodological infrastructure for the study, covering the datasets, computational architecture, and experimental protocols designed for a reproducible and robust comparative analysis.

4.1.1. Datasets

This study uses two image datasets, FIN-Benthic2 and BIOSCAN-1M, that represent the inherent class imbalance challenges in ecological classification. These datasets were selected for their long-tailed distributions and taxonomic diversity, providing distinct contexts to assess the generalizability of mitigation strategies.

FIN-Benthic2 dataset (Ärje et al., 2020; Raitoharju, Ärje, et al., 2018) comprises multi-image data of freshwater macroinvertebrates with hierarchical taxonomic labels. The dataset was collected using a dual-camera imaging system. The subset employed for this comparison consists of 460,004 images, representing 39 distinct taxa (a maximum of 50 images per specimen). Classification was performed using the most specific available taxon label for each specimen (see Table A.10), corresponding to these 39 categories. The dataset exhibits significant class imbalance, with the number of images per taxon ranging from 490 in the smallest class to 44,240 in the largest.

BIOSCAN-1M Insect dataset (Gharaee et al., 2023) provides approximately one million images taxonomically classified to the family level by experts. A key characteristic of this dataset is its inherent long-tailed class distribution, which results in significant imbalance across families and makes it highly relevant for this research. For the comparative analysis, a working subset was derived from the collection, comprising 1,106,784 instances from 192 families, selected based on a minimum representation of 100 images per family. This subset exhibits pronounced imbalance, with the majority class containing 259,104 instances and the minority class only 101. The curated subset retains both the broad taxonomic scope and the challenging imbalance characteristics of the original dataset, rendering it suitable for this evaluation.

4.1.2. Experimental design overview

We conducted a structured experimental investigation to evaluate class imbalance mitigation strategies. We first established baseline performance by evaluating three CNN architectures, AlexNet (Krizhevsky et al., 2012), MobileNetV2 (Howard et al., 2017), and EfficientNetB0 (Tan & Le, 2019), via transfer learning (Section 4.2.1). These models

were chosen for their prevalence in prior research and diverse architectural principles. Based on the initial results, we selected EfficientNetB0 for all subsequent experiments.

First, we assessed algorithm-level methods by comparing three loss functions: CE loss, Focal loss, and CI loss, to evaluate their effectiveness in addressing class distribution disparities (Section 4.2.2). Subsequently, we evaluated data-level methodologies. We began by examining data augmentation techniques to measure their impact on minority class representation (Section 4.2.3). We then investigated synthetic data generation using a DCGAN and other recent GAN models to investigate its feasibility for creating artificial samples (). Finally, we conducted a comparison of sampling strategies including upsampling, downsampling, and a combined approach, to analyze their effects on data distribution (). In hybrid methods, we used bagging-based ensemble learning, where we integrated previously evaluated strategies to assess potential performance improvements (Section 4.2.4).

Throughout these experiments, model performance was assessed using metrics suitable for imbalanced classification (Section 3.3), including Macro F1-score, Balanced Accuracy (BACC), and Geometric Mean (G-mean). Also overall Accuracy and Matthews Correlation Coefficient (MCC) values are reported. This progressive design allowed for a systematic evaluation of each mitigation strategy's effectiveness on fine-grained taxonomic classification tasks with severe imbalance.

4.1.3. Implementation details

The experimental implementation employs a standardized computational framework designed to ensure reproducibility and methodological consistency across all evaluated strategies. We have made the complete implementation details and source code publicly available[1] to facilitate replication and extension of this research. For computations, we used the Puhti CSC supercomputer[2] equipped with NVIDIA V100 GPUs.

We selected the PyTorch Lightning framework for its robust support of distributed training and experimental reproducibility, enabling comparison across diverse methodological configurations. We sourced all pretrained models used in this research from the timm (PyTorch Image Models) library,[3] which provides standardized implementations of state-of-the-art architectures pretrained on ImageNet. This ensures consistency in baseline model initialization and facilitates reproducible transfer learning experiments.

Based on preliminary experimentation and established best practices, we set the key implementation parameters as follows: a fixed learning rate of 0.002 with AdamW (Loshchilov & Hutter, 2017) optimization, a batch size of 256, and 50 training epochs to ensure convergence. We standardized inputs to a 224×224 pixel resolution to maintain consistency with pretrained model expectations. We decided to allow full network adaptation during transfer learning, rather than freezing early layers, to reflect the significant domain shift between natural images and microscopic taxonomic imagery. These implementation choices prioritize experimental validity through standardized protocols while maintaining computational feasibility.

To ensure a fair comparison, hyperparameters for both Focal and CI losses were systematically tuned using a grid search on a separate validation set for each dataset. For Focal loss, we utilized the α-balanced variant, where α was dynamically calculated based on the inverse class frequency of each dataset to explicitly address imbalance. The focusing parameter γ was set to 2.0 after evaluating a range of values ($\gamma \in \{0.5, 1, 2, 5\}$), following the standard provided by Lin et al. (2017). For the CI loss, hyperparameters k and θ were selected via a grid search. For the 39-class FIN-Benthic2 dataset, we used $k = 0.3$ and $\theta = 3.0$. Conversely, for the 192-class BIOSCAN dataset, the default parameters of $k = 1.0$ and $\theta = 0.5$, as established in the original formulation (Yue et al., 2023), provided the best performance.

For the DCGAN experiments, we trained the model using a batch size of 128, a noise dimension of 100, 64 feature maps, and a learning rate of 2×10^{-4} for both the generator and discriminator. For the WGAN experiments, we used a batch size of 64, latent dimension of 128, learning rate of 5×10^{-5} with RMSprop, 5 critic iterations per generator step, and weight clipping of 0.01. Training ran for 50 epochs at 64×64 resolution. For the ProGAN experiments, we progressively grew the model from 4×4 to 128×128 with 30 epochs per stage. We used a latent dimension of 256, learning rate of 10^{-3} with Adam ($\beta_1 = 0$, $\beta_2 = 0.99$), WGAN-GP (WGAN with gradient penalty) loss ($\lambda = 10$), and batch sizes of 32–16 depending on resolution. For the StyleGAN3 experiments, we used the translation-equivariant configuration (stylegan3-t) at 256×256 with a batch size of 8, R1 regularization $\gamma = 8.2$, ADA augmentation, conditional training on 192 insect families, and trained for 4000 kimg (thousands of images shown to the discriminator during training).

Table 2
Performance of ImageNet pretrained architectures on FIN-Benthic2 data (EfficientNet* = no pre-training)

Metrics	AlexNet	EfficientNet	Mobile-VNet
F1-score	0.7258	**0.8542**/*0.7415**	0.8293
G-mean	0.8470	**0.9215**/*0.8559**	0.9029
BACC	0.7207	**0.8510**/*0.7358**	0.8175
MCC	0.8220	**0.9174**/*0.8275**	0.8970
Accuracy	0.8292	**0.9207**/*0.8345**	0.9012

4.2. Comparative results

In this subsection, we present the empirical findings from our evaluation of various class imbalance mitigation strategies. We report the results in a structured manner, starting with the baseline model performance and followed by the outcomes of algorithm-level, data-level, and hybrid method evaluations. The performance is reported using quantitative metrics selected for imbalanced classification scenarios, detailing the effectiveness of each strategy across different taxonomic granularities.

4.2.1. Baseline model selection

To establish a baseline, we evaluated three CNN architectures using transfer learning on the FIN-Benthic2 dataset: AlexNet (Krizhevsky et al., 2012), EfficientNetB0 (Tan & Le, 2019), and MobileNetV2 (Howard et al., 2017). The experimental results are presented in Table 2. EfficientNetB0 achieved the highest performance across all metrics, including F1-score and BACC, outperforming both AlexNet and MobileNetV2. We also compared the transfer learning approach to training EfficientNetB0 from scratch, and the pretrained model showed a notable performance advantage.

Based on these results, we selected EfficientNetB0 as the base architecture for all subsequent experiments evaluating class imbalance mitigation strategies. Consistent performance patterns were also observed on the BIOSCAN dataset. The primary goal of this step was not an architectural comparison but to select a strong, representative model for the core analysis. The confusion matrix and per-class results including true positives, support, precision, recall, and F1-scores are given in supplementary materials (Tables A.7–A.8).

4.2.2. Experimental results for algorithm-level methods

We evaluated three algorithm-level approaches by testing the performance of CE loss, Focal loss, and CI loss. We trained the EfficientNetB0 model with each loss function on both the FIN-Benthic2 and BIOSCAN datasets.

[1] https://github.com/mskhan793/taxonomist/tree/imbalance
[2] https://www.puhti.csc.fi
[3] https://huggingface.co/timm

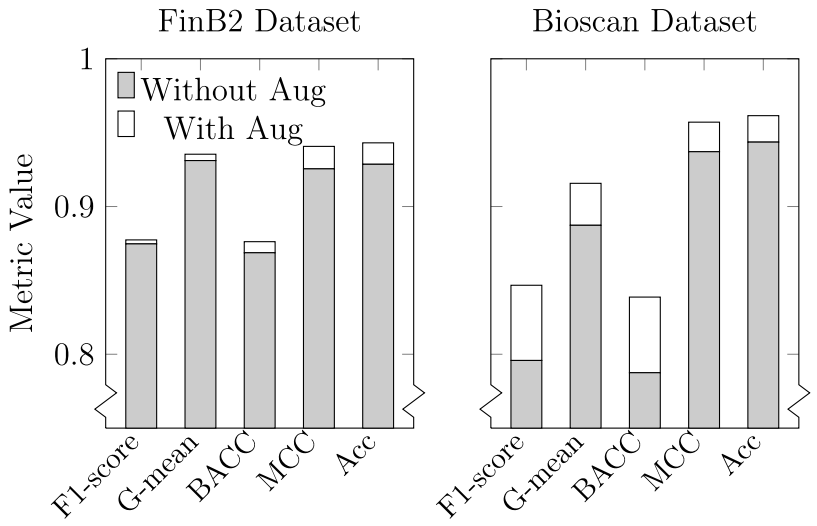


Fig. 2. Impact of data augmentation (Aug) on classification performance for FinB2 (left) and Bioscan (right) datasets using CI loss.

Table 3
Comparative performance of CE loss, Focal loss, and CI loss on FIN-Benthic2 and BIOSCAN datasets.

Metrics	FIN-Benthic2			BIOSCAN		
	CE loss	Focal loss	CI loss	CE loss	Focal loss	CI loss
F1-score	0.8542	0.8469	**0.8747**	0.7926	0.6534	**0.7958**
G-mean	0.9215	0.9201	**0.9311**	0.8829	0.8714	**0.8873**
BACC	0.8510	0.8485	**0.8687**	0.7798	0.7600	**0.7875**
MCC	0.9174	0.9150	**0.9255**	**0.9416**	0.8117	0.9371
Accuracy	0.9207	0.9184	**0.9286**	**0.9477**	0.8282	0.9436

The results, detailed in Table 3, show that CI loss consistently outperforms the other functions on both datasets across metrics designed for imbalanced classification, i.e, F1-score, G-mean, and BACC. On the FIN-Benthic2 dataset, CI loss achieved the highest scores across all metrics, but the differences between the losses were relative small. On the BIOSCAN dataset, the differences were more pronounced: CI loss maintained strong performance, whereas Focal loss performed poorly, particularly failing to classify uncommon species. While standard CE loss achieved highest accuracy, its performance on minority taxa was lower, as indicated by the imbalance-specific metrics.

4.2.3. Experimental results for data-level methods

This subsection evaluates data-level strategies, data augmentation, sampling, and synthetic generation to address class imbalance, assessing their impact on underrepresented taxa and majority class performance.

Data Augmentation:

We evaluated the effect of data augmentation on both the FIN-Benthic2 and BIOSCAN datasets, using the CI loss as the base loss function. For each image in the training set, we dynamically applied a unique combination of pixel-level (e.g., ColorJitter, GaussianBlur) and geometric (e.g., Affine, Rotate) transformations at every epoch. The results, shown in Fig. 2, varied between the datasets. On FIN-Benthic2 dataset, augmentation yielded only a modest improvements across all metrics. We observed a slight increase in the F1-score, G-mean, and BACC, with a corresponding gain in overall accuracy. Conversely, for BIOSCAN, augmentation led to more substantial gains, particularly in the metrics sensitive to imbalance. The F1-score and BACC increased significantly, along with a notable improvement in the G-mean.

Synthetic Data Generation:

As part of our evaluation of data-level methods for class imbalance, we tested multiple generative models, including DCGAN, WGAN, ProGAN, and StyleGAN3, on the BIOSCAN dataset. Fig. 3 shows a qualitative comparison between real and GAN-generated insect images. The visual quality improved progressively across architectures, with DCGAN producing the weakest samples and StyleGAN3 generating the most visually plausible images.

To further evaluate whether visually plausible synthetic images can improve minority class classification, we selected two families, *Cydnidae* and *Sepsidae*, for targeted augmentation using StyleGAN3. Fig. 4 shows a visual comparison of real and StyleGAN3-generated images for the two selected minority classes. We generated 1000 synthetic images for each class and added them to the training set, without applying any additional regularization or augmentation, in order to isolate the effect of synthetic data alone. For *Cydnidae*, the number of training samples increased from 41 to 1041, and for *Sepsidae*, from 241 to 1241. Fig. 5 shows the quantitative impact of data augmentation via StyleGAN3. The added synthetic data did not improve classification performance. For *Cydnidae*, recall dropped, while for *Sepsidae*, recall remained unchanged. Fig. 5 further shows that the synthetic augmentation reduced precision, recall, and F1-score for *Cydnidae*, while producing only marginal changes for *Sepsidae*. These results indicate that, although newer GAN architectures produce visually plausible images, the generated samples did not provide a consistent benefit for downstream minority class classification in our experiments.

Sampling Techniques:

To evaluate data-level sampling techniques, we implemented three distinct strategies on both the FIN-Benthic2 and BIOSCAN datasets: upsampling (US), downsampling (DS), and a combined up–down-sampling (US-DS) approach.

For the FIN-Benthic2 dataset, the number of images in training data ranged from 375 in the smallest class to 28,139 in the largest. We established a sampling threshold of 12,000 images per class. To achieve this, upsampling was applied to minority classes by duplicating images, while downsampling was applied to majority classes by randomly removing images. This procedure reduced the original imbalance ratio. For instance, the ratio improved from 75.04:1 to 2.34:1 with upsampling and to 32:1 with downsampling.

For the BIOSCAN dataset, which exhibits an even more extreme imbalance (from 41 to 115,406 images per family in training data). We selected the mean number of samples per class in the training data, 2562, as the threshold for sampling. Applying the same upsampling and downsampling procedures reduced a representative imbalance ratio from 2814.78:1 to 45.05:1 and 62.49:1, respectively. The combined



Fig. 3. Comparison of real and GAN-generated insect images. The first row shows real images, followed by images generated by DCGAN (row 2), WGAN (row 3), ProGAN (row 4), and StyleGAN3 (row 5).



Fig. 4. Comparison of real and StyleGAN3-generated images for the two selected classes. Row 1: real *Sepsidae*; row 2: StyleGAN3-generated *Sepsidae*; row 3: real *Cydnidae*; row 4: StyleGAN3-generated *Cydnidae*.

up–down-sampling strategy involved applying both upsampling to minority classes and downsampling to majority classes to meet the target threshold.

We applied a strong stochastic augmentation dynamically generating a unique combination of pixel-level (e.g., ColorJitter, GaussianBlur) and geometric (e.g., Affine, Rotate) transformations at every epoch. This stochastic augmentation ensures that when the model sees duplicated samples, each one looks different (is augmented differently) in every epoch. We used CI loss as the baseline for all experiments to see the effect of sampling. The results are illustrated in Figs. 6 and 7.

On FIN-Benthic2 dataset, the combined up–down-sampling strategy yielded the most balanced performance, improving the G-mean and BACC with a slight increase in overall accuracy. In contrast, upsampling alone reduced the F1-score and BACC, while downsampling achieved the highest overall accuracy but performed less effectively on imbalance-specific metrics.

For BIOSCAN data, upsampling resulted in significant improvements across all metrics, including accuracy, F1-score, G-mean, and BACC. Conversely, downsampling led to a slight decrease in overall accuracy, though it improved some imbalance-oriented metrics. The combined up–down-sampling strategy also improved imbalance metrics but at the cost of lower overall accuracy compared to the baseline.

To assess whether the observed improvement was statistically significant, we selected Macro F1-score and performed a Wilcoxon

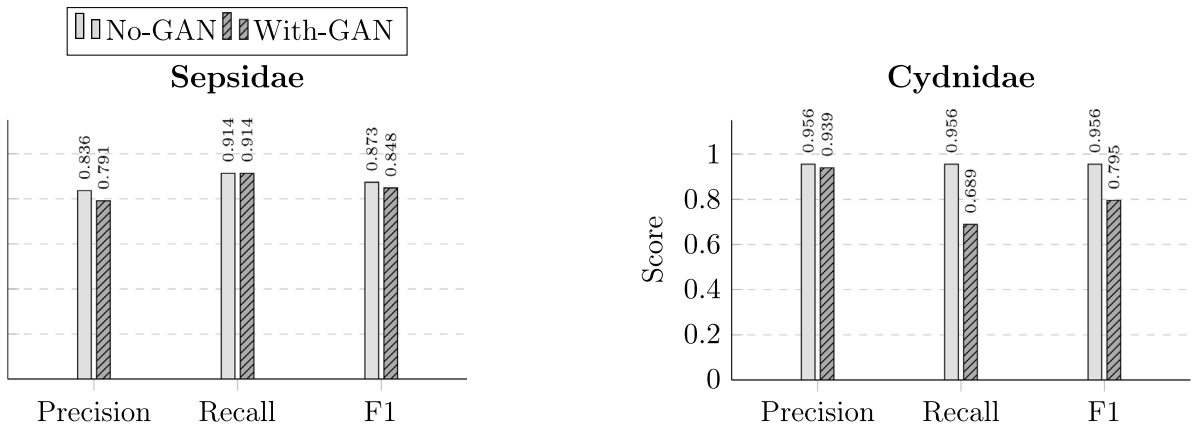


Fig. 5. Precision, recall, and F1 for Cydnidae and Sepsidae before and after adding StyleGAN3-generated training images. Light bars: No-GAN baseline; dark hatched bars: With-GAN.

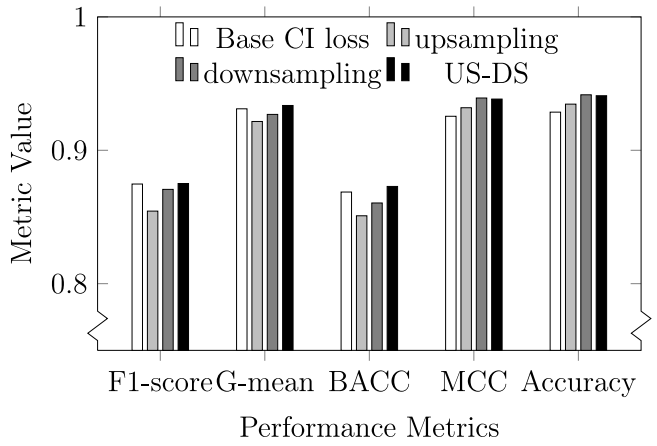


Fig. 6. Performance of upsampling, downsampling, and combined up–down-sampling (US-DS) strategies using CI loss on FIN-Benthic2 dataset.

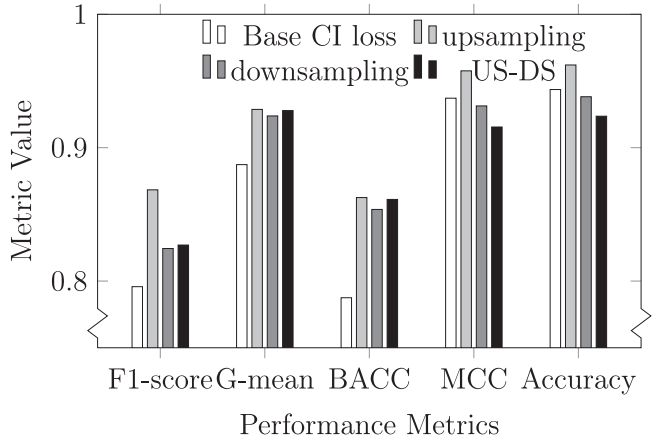


Fig. 7. Performance of upsampling, downsampling, and combined up–down-sampling (US-DS) strategies using CI loss on BIOSCAN dataset.

signed-rank test using 12 paired observations obtained from 3 mutually exclusive folds and 4 random seeds. The result shows that the improvement of upsampling compared to the baseline was statistically significant ($p = 0.000488$). The paired observations and additional Wilcoxon test details are provided in the supplementary material (Tables A.15 and A.16).

4.2.4. Experimental results for hybrid methods through ensemble learning

We evaluated a hybrid approach using a bagging-based ensemble. We divided each dataset into three subsets with replacement (bootstrap aggregating) and trained three separate models. We then combined their predictions using an averaging pooling strategy and compared the ensemble's performance against the baseline single-model implementation. The main hybrid-method trends for the F1-score and G-mean are illustrated in Figs. 8 and 9, while full numerical results are provided in the supplementary material (Table A.5).

For the FIN-Benthic2 dataset, we tested bagging with different data strategies. As shown in Fig. 8, the basic Bag-CI ensemble (bagging with CI loss) significantly improved the F1-score and G-mean. This upward trend was consistent across all evaluation metrics, as detailed in Table A.5. The Bag-CI-Aug method, which integrated data augmentation, achieved the highest scores across all metrics. The Bag-CI-US-DS (combining up–down-sampling and CI loss with bagging) method also outperformed its non-bagged counterpart and the baseline. Detailed results including confusion matrices and per-class results for Bag-CI-Aug approach are available in the supplementary material (Tables A.9–A.12).

To verify that the improvement of the best hybrid method for FIN-Benthic2 was statistically significant, we additionally applied a paired bootstrap test on our predictions. This test repeatedly resamples the matched predictions with replacement and evaluates whether the observed performance gain remains consistent across the resampled sets. The result confirmed that the improvement of the Bag-CI-Aug method over the baseline was statistically significant. Additional statistical details are provided in the supplementary material (Table A.16).

On the BIOSCAN dataset, ensemble methods also yielded significant improvements (Table A.6). Fig. 9 visually plots the F1-score and G-mean improvements for each method. The Bag-CI outperformed its non-bagged version but the improvement was less noticeable for G-mean and BACC. Bag-CI-Aug delivered stronger results than the basic Bag-CI ensemble, with clear enhancements in the F1-score and BACC. The Bag-CI-US method produced the best results, achieving the highest scores across all evaluation metrics. This method incorporated upsampling, which we previously identified as the most effective sampling strategy for this dataset. Supplementary material (Tables A.13 and A.14) provide comprehensive classification metrics stratified by support size. Specifically, Table A.13 details the performance of the baseline CI loss model, while Table A.14 documents the enhanced performance achieved by the best performing method for the BIOSCAN dataset: a hybrid approach combining Bagging, Upsampling, and CI loss.

To verify that the improvement of the best hybrid method was statistically significant, we additionally applied a paired bootstrap test on

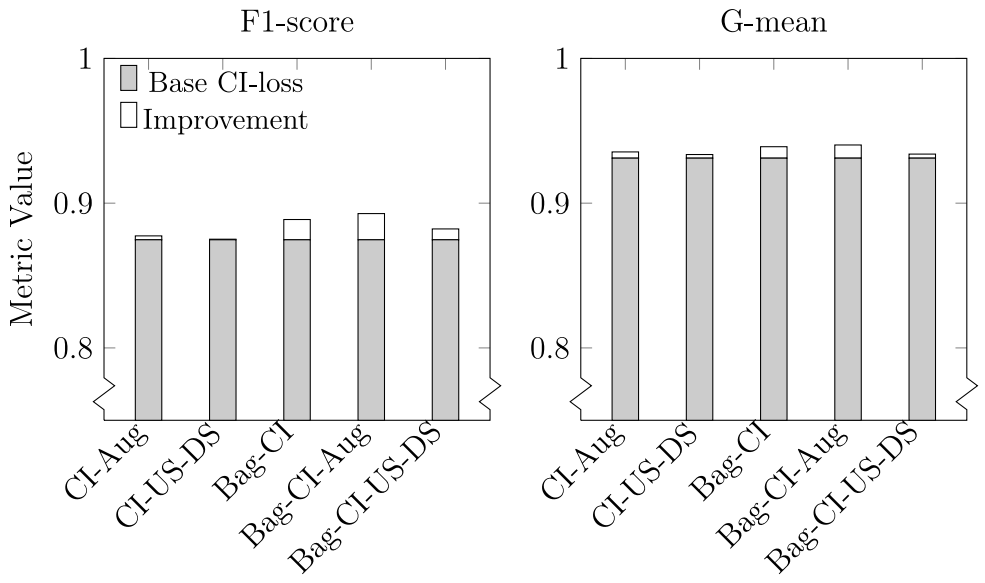


Fig. 8. Hybrid ensemble performance breakdown for the FIN-Benthic2 dataset, showing F1-score (left) and G-mean (right) improvements over the base CI-loss model.

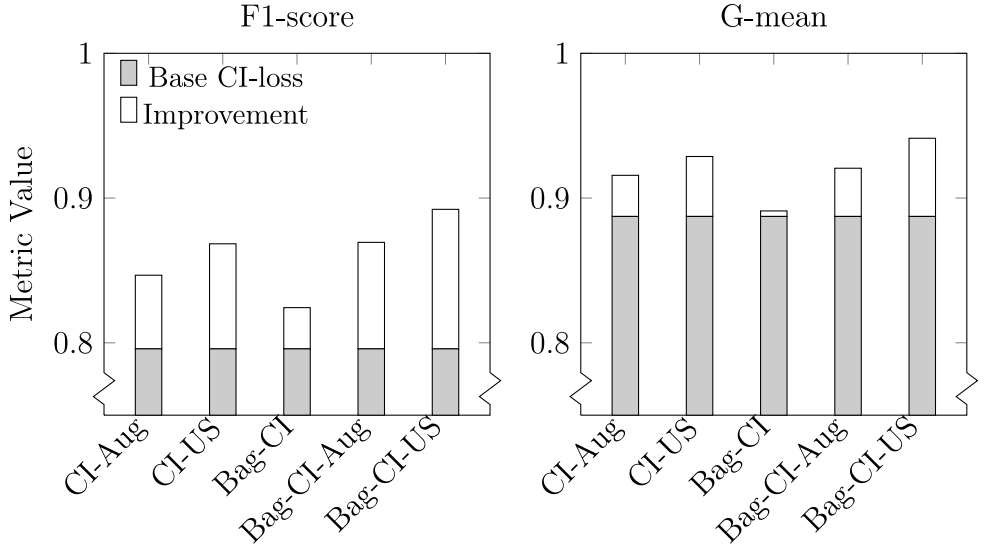


Fig. 9. Hybrid ensemble performance breakdown for the BIOSCAN dataset, showing F1-score (left) and G-mean (right) improvements over the base CI-loss model.

the BIOSCAN predictions. The result confirmed that the improvement of the Bag-CI-US method over the baseline was statistically significant. Additional statistical details are provided in the supplementary material (Table A.16).

4.2.5. Misclassification analysis

We also examined the most frequent misclassifications in the FIN-Benthic2 dataset. The per-class statistics are provided in the supplementary material (Tables A.7–A.10). We observed two dominant confusion patterns that reflect the difficulty of discrimination at different levels of the taxonomic hierarchy. Taxonomic confusion is relatively easier to avoid at higher levels such as order and family compared to finer levels such as genus and species, where morphological differences become increasingly subtle and closely related organisms look increasingly similar, making correct identification harder. Our results follow a similar pattern.

Figs. 10 and 11 show focused confusion matrices for the eight most frequently confused taxa under the baseline and best-performing method, respectively. The matrices are organized by taxonomic hierarchy, grouping taxa by order (solid boundary lines) and family (dashed boundary lines), so that the taxonomic level of each confusion is easily visible. The most frequent confusions occur at the genus and species level, where close relatives are morphologically near-identical. Within the genus *Polycentropus*, the rarer *P. irroratus* (ID 7; 37 individuals, 1817 training images) is confused with *P. flavomaculatus* (ID 20; 150 individuals, 7381 training images), which supplies over four times more training data. The same pattern holds for *Leuctra nigra* (ID 4; 15 individuals, 750 training images) being predicted as *Leuctra* sp. (ID 25; 192 individuals, 9600 training images), and *Baetis vernus* group (ID 14; 109 individuals, 5315 training images) being predicted as *Baetis rhodani* (ID 33; 261 individuals, 12,794 training images). In each case confusion is skewed toward the more abundant close relative, a direct consequence of class imbalance amplifying an already difficult morphological discrimination task.

The second pattern is noteworthy because it occurs across family boundaries rather than within a genus. *Capnopsis schilleri* (ID 2, Capniidae; 15 individuals, 750 training images) is misclassified as *Amphinemura borealis* (ID 28, Nemouridae; 209 individuals, 10,450

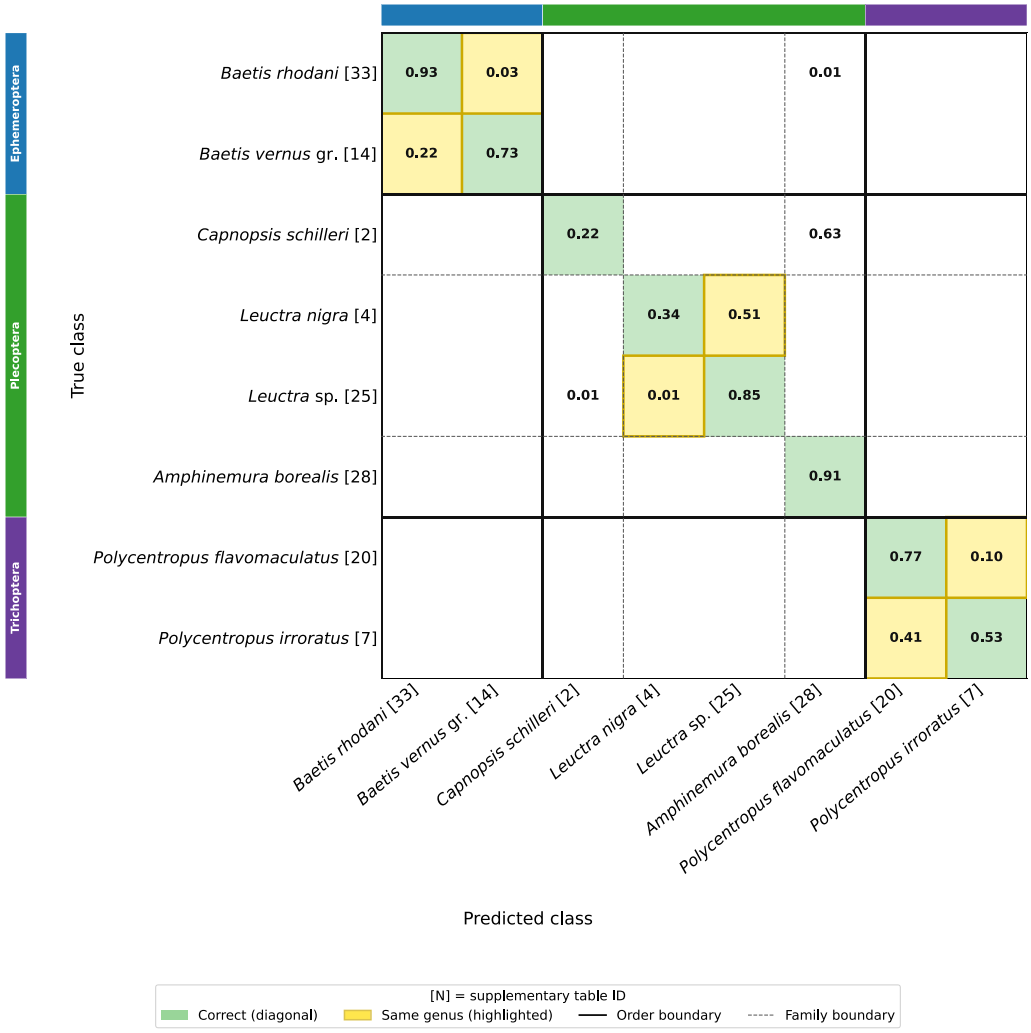


Fig. 10. Focused confusion matrix for the eight taxa involved in the most frequent misclassifications (FIN-Benthic2, baseline CI loss). Green: correct predictions; yellow: within-genus confusion. Solid lines denote order boundaries; dashed lines denote family boundaries. Numbers in brackets are supplementary table IDs. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

training images) at a rate of 63% on the test data, which is higher than any of the genus-level confusion pairs. With 14 times more training images and 14 times more unique individuals, *Amphinemura borealis* overwhelms the model's capacity to learn the rare *Capnopsis schilleri*, demonstrating that class imbalance can drive misclassification across taxonomic family boundaries and not only within a genus. Fig. 12 illustrates both patterns by displaying representative specimen images for each confused pair. The visual similarity within each pair, particularly at the genus level, demonstrates that the observed confusions are not model artifacts but reflect genuine morphological ambiguity that is further compounded by class imbalance.

In the best-performing method (Bag-CI-Aug), *Capnopsis schilleri* improved substantially, from 63% to 24% misclassified as *Amphinemura borealis*. However, the within-genus minority species worsened: *Leuctra nigra* recall dropped from 34% to 11%, with 88% of its predictions absorbed by *Leuctra* sp., and *P. irroratus* recall dropped from 53% to 20%, with 68% absorbed by *P. flavomaculatus*. Overall, the method reduced cross-family confusion while within-genus confusion increased for the minority species in each pair.

5. Discussion

In this paper, we provide a comprehensive comparative empirical analysis of class imbalance mitigation strategies for image-based ecological fine-grained taxa identification tasks that naturally have very high class imbalance ratios. Our experimental analysis demonstrates that a multi-level strategy can be highly effective for taxonomic classification on imbalanced datasets. We find that establishing a strong baseline model is a critical first step, but its performance is substantially enhanced by implementing targeted class imbalance mitigation strategies at the algorithmic, data, and hybrid levels. The consistent performance gains across these interventions underscore that no single method serves as a universal solution; instead, a tailored combination of techniques achieves the most robust results.

A key aspect of our work is the critical importance of metric selection in evaluating model performance. Standard accuracy is a misleading metric in imbalanced scenarios, as it often masks poor performance on rare taxa. In contrast, we show that metrics designed for imbalance, such as the macro-averaged F1-score, G-mean, and BACC, provide a much more reliable assessment of a model's ability to classify both rare and common species. Based on these findings, we provide a metric selection guide (Table 4) to help practitioners choose appropriate evaluation criteria for their specific research goals.

Each of these metrics provides a unique lens for evaluating performance on imbalanced data. The F1-score is crucial in scenarios where both precision and recall are significant, as it ensures a balance between the two. The G-mean is particularly sensitive to poor performance in any single class; a low recall for even one class will drastically

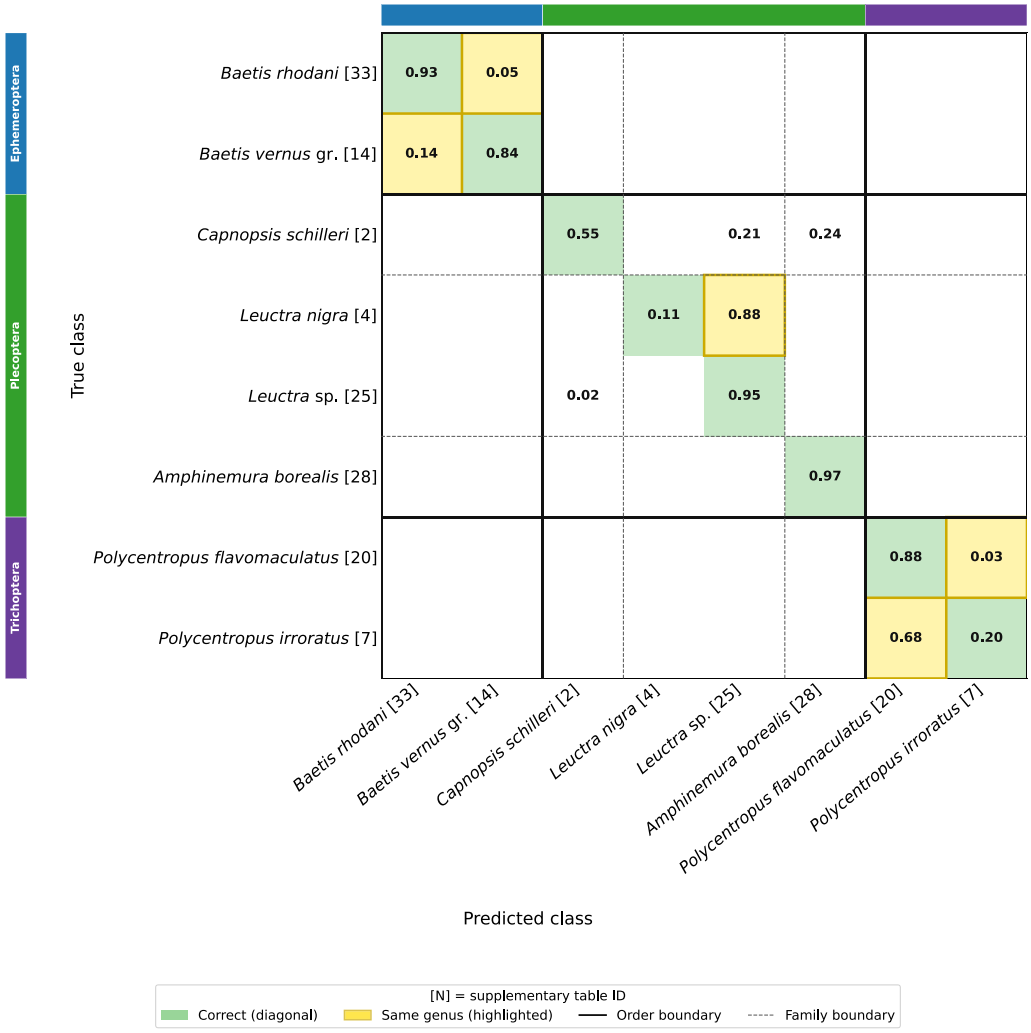


Fig. 11. Focused confusion matrix for the same eight taxa under the best-performing method (Bag-CI-Aug). Green: correct predictions; yellow: within-genus confusion. Solid lines denote order boundaries; dashed lines denote family boundaries. Numbers in brackets are supplementary table IDs.

Table 4
Metric selection guide for imbalanced classification tasks.

Task goal	Primary metric(s)	Secondary metric(s)	Rationale
Default for High Imbalance	BACC	Macro-F1, G-mean	Robust baseline for imbalance; unweighted average of per-class recall
Penalize failure on any class	G-mean	Macro-F1, BACC	Strictest "macro" metric; collapses to 0 if any single class has 0 recall
Equal Class Importance	BACC, Macro-F1	G-mean, MCC	Fair evaluation across classes; mitigates majority bias
Overall Instance Performance	Accuracy	All	Per sample correct predictions; useful if data is balanced

lower the score, making it an excellent indicator of balanced classification. Similarly, BACC is beneficial for multi-class problems because it averages the recall of each class, giving equal importance to the correct prediction of both rare and common taxa. By averaging per-class performance, BACC offers a key metric for monitoring a model's generalization ability across an imbalanced dataset. It is also a good idea to always use secondary metrics. The differences between the metrics may reveal important insights into the model behavior.

Our evaluation of algorithm-level strategies shows that the choice of loss function is critical for handling imbalanced taxonomic data. We find that CI loss consistently outperforms the commonly used CE loss and Focal loss across both datasets. Its success stems from its dynamic weighting mechanism, which assigns higher priority to rare classes based on their scarcity while simultaneously reducing penalties for well-classified samples. This adaptive approach, which considers both class rarity and model confidence during training, proves to be highly effective.

In contrast, other loss functions show significant limitations. The standard CE loss provides a functional baseline but lacks the discriminative power needed for underrepresented taxa. Focal loss proves to

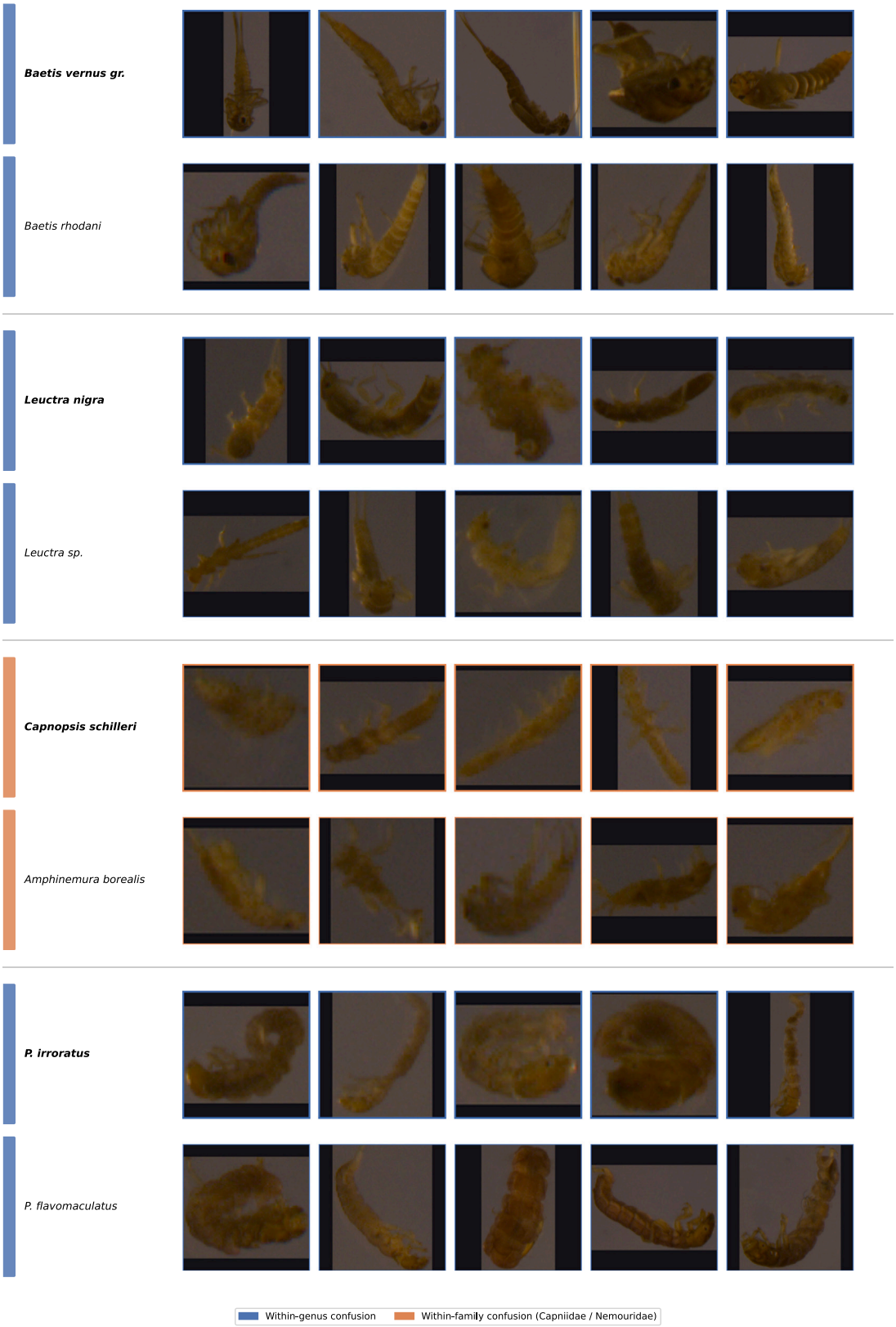


Fig. 12. Representative specimen images for the most frequent confusion pairs (five images per species; blue borders: within-genus pairs; orange borders: within-family pair). See text for details. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

be dataset-sensitive, performing poorly on the extreme imbalance of the BIOSCAN dataset, where fixed focusing parameters can suppress gradients for minority classes. These findings underscore a critical conclusion: for ecological tools to reliably identify rare species, the loss function must dynamically balance class importance. CI loss achieves this effectively, demonstrating its reliability across different levels of

data imbalance. CI loss has not been commonly used in ecological imbalanced classification tasks, but our results suggest that practitioners should consider changing it in their future studies.

The analysis of data-level methods shows noticeable, dataset-specific patterns. For FIN-Benthic2 dataset, data augmentation yielded only marginal improvements, whereas for BIOSCAN dataset, the gains were substantial. This is likely because FIN-Benthic2 contains multiple images per specimen, which provides a 'natural augmentation' and variability, while with BIOSCAN dataset data augmentation can increase the image variability more.

In contrast, our generative augmentation experiments revealed important limitations. While DCGAN produced clear morphological degradation in the generated samples, experiments with WGAN, ProGAN, and StyleGAN3 demonstrated that visual quality can be improved substantially with more advanced architectures. Among the tested models, StyleGAN3 produced the most realistic synthetic insect images, including samples that were visually difficult to distinguish from real ones. However, even these visually strong synthetic images did not provide consistent gains in downstream minority class classification. In the targeted StyleGAN3 experiments on BIOSCAN, augmentation did not improve the selected minority classes and even reduced recall for *Cydnidae*. This suggests that visual realism alone is not sufficient; synthetic samples must also preserve the discriminative taxonomic cues required by the classifier. Therefore, our results should not be interpreted as a general failure of generative augmentation, but rather as evidence that current GAN-based synthesis, even with stronger modern architectures, remains challenging for fine-grained ecological classification in class-imbalanced settings. At the same time, these findings should not be generalized to more recent methods or to diffusion-based generative models, since these were outside the scope of the present study.

Sampling strategies also produced different results for each dataset. For FIN-Benthic2, the combined up–down-sampling method showed a marginal improvement in imbalance metrics while increasing accuracy. For BIOSCAN, however, upsampling provided significant improvements across all metrics. This variation highlights the importance of dataset structure: minority classes in FIN-Benthic2 have many images but of fewer unique specimens, so simple upsampling is less helpful. In training data, the rarest FIN-Benthic2 classes comprise as few as 12–15 unique individuals contributing 600–750 images (50 images per individual), meaning that augmentation and upsampling operate on the same small pool of specimens, providing limited gain in diversity and increasing the risk of overfitting. BIOSCAN has more unique specimens per class (one image each), with the smallest training classes containing 41–57 unique individuals. In this imbalanced setting, augmentation and sampling are more helpful for BIOSCAN because they reduce overfitting by exposing the model to greater variation during training. A second structural difference is the taxonomic classification level: 38 out of 39 FIN-Benthic2 taxa are at genus or species level, whereas BIOSCAN labels are at the family level. As shown in the misclassification analysis (Section 4.2.5), visual discrimination becomes progressively harder at lower taxonomic levels, making it harder for any single mitigation strategy to fully recover minority class performance in FIN-Benthic2. These findings show that the results should be interpreted across all compared methods rather than only against the baseline, and they further underscore that dataset-specific sampling strategies are necessary to tackle the challenges of class imbalance in ecological classification. To maintain experimental consistency, we applied a uniform augmentation policy across all classes. However, this may not be equally optimal for every rare taxon, which highlights the trade-off between a consistent experimental baseline and the possible need for class-specific augmentation tuning.

Our experimental findings for hybrid methods demonstrate two principal outcomes. First, ensemble implementations consistently surpassed their single-model baselines across both datasets, with BIOSCAN exhibiting more substantial improvements in imbalance-sensitive metrics compared to FIN-Benthic2. Second, the integration of data-level and algorithm-level strategies within these ensemble frameworks yielded dataset-specific optimal configurations. For FIN-Benthic2 dataset, the Bag-CI-Aug method achieved optimal performance. BIOSCAN dataset showed its strongest results with the Bag-CI-US method, which excelled in F1-score and BACC by directly amplifying its rare families. This divergence in optimal strategies directly corresponds to the fundamental dataset structures identified in our data-level analysis. While FIN-Benthic2 displayed modest gains across evaluation criteria, BIOSCAN revealed significant enhancements in metrics designed for imbalanced cases, as visually highlighted by the F1-score and G-mean improvements in Fig. 9. The superior performance of these dataset-tailored ensemble configurations underscores the value of empirically-guided strategy selection rather than the universal application of a single hybrid approach in ecological classification contexts.

Evaluating our observations in the light of methods currently applied in ecological fine-grained taxa identification, we note that some prior biodiversity studies (Bjerge et al., 2023; Goldsworthy & Matam, 2021) have reported successful use of Focal loss, but in our case it produced comparable performance on one dataset and very weak performance on the other dataset. This suggests that different loss functions should be explored depending on the target application. CI loss has not been used in prior ecological studies, but should not be omitted in future studies.

Many studies have reported successful use of sampling methods (Binta Islam et al., 2023; Mohammadisabet et al., 2025; Simões et al., 2023), and in our experiments we also observed positive effects. However, for some classes these strategies reduced performance and increased the risk of overfitting. For minority classes such as *Leuctra nigra*, we observed a severe degradation in performance (see Section 4.2.5). In such cases, class-specific augmentation or sampling strategies omitted in our study may be more suitable and should be further evaluated in future studies.

While prior related studies have not commonly considered ensemble approaches, our results encourage exploration of more advanced ensemble methods, including deep learning based ensembles. At the same time, ensemble-based methods have a practical limitation in their increased computational cost compared to single-model solutions. In this work, we used three models in the ensemble as a balance between improved performance and computational cost, and probability averaging was adopted as a simple fusion strategy. Therefore, while the ensemble methods achieved the best performance, lighter approaches may still be more practical in resource-constrained deployment settings.

In general, we suggest that more focus should be placed on the effect of class imbalance itself, and on clearly understanding how each method helps to alleviate this problem. Our results are in line with previous reported observations in general imbalance mitigation surveys (e.g., Carvalho et al. (2025), W. Chen et al. (2024)) showing that no single method provides superior performance in all cases, and that a careful combination of different strategies should be considered.

The two datasets considered in this study differ in their taxonomic groups and levels, which affects both classification difficulty and ecological interpretation. In general, moving to lower taxonomic levels, such as genus or species makes the identification task more challenging (see also Section 4.2.5), because differences between taxa are more fine-grained and in many cases their manual identification requires microscopic analysis. In addition, the class imbalance often becomes more severe at lower taxonomic levels. However, class similarity and imbalance do not depend only on the taxonomic level, but also on the specific taxonomic groups and taxa present at the dataset sampling cite. Even our experiments on FIN-Benthic2 dataset (see Table A.10) show that in some cases better identification performance can be obtained for taxa at lower taxonomic levels compared to those at higher levels. For this reason, our conclusions should be interpreted mainly in the

context of fine-grained ecological classification under severe imbalance irrespective of the taxonomic rank of the classes.

Our experimental findings indicate several promising future research directions for taxonomic classification in imbalanced datasets. The development of specialized architectures focusing on minority class feature extraction could enhance classification reliability in taxonomic tasks. Investigation of customized loss functions and their combinations presents opportunities for improved learning in imbalanced scenarios. For generative augmentation, our results show that although advanced GANs improve the visual quality of synthetic insect images, they still do not provide consistent gains for downstream minority class classification in fine-grained class-imbalanced setting. This indicates that future work should focus not only on visual realism, but also on preserving the discriminative taxonomic cues required by the classifier. In this regard, diffusion-based models, few-shot conditional generation, and recent long-tail recognition methods may offer more suitable directions for ecological image synthesis. Moreover, research could explore auto-selection strategy-based hybrid methods, where the advantages of different approaches can be combined. Additionally, exploration of ensemble methodologies with dynamic weighting mechanisms based on class distributions offers potential for enhanced taxonomic classification performance.

CRediT authorship contribution statement

Muhammad Shahzad Khan: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing,. **Eero Siivola:** Conceptualization, Methodology, Validation, Writing – review & editing, Supervision. **Jenni Raitoharju:** Conceptualization, Methodology, Validation, Writing – review & editing, Supervision, Funding acquisition.

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Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Paperpal in order to improve language and readability. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Supplementary material

Table A.5 summarizes the evaluation of hybrid ensemble strategies for the FIN-Benthic2 dataset using F1-score, geometric mean (G-mean), BACC, MCC, and accuracy metrics. These results compare the baseline CI loss model with augmentation, up–down-sampling, and bagging-based hybrid strategies. Table A.6 summarizes the evaluation of hybrid ensemble strategies for the BIOSCAN dataset using the same aggregate metrics. The table shows the comparative effect of augmentation, up-sampling, and bagging-based hybrid methods relative to the baseline CI loss model.

Table A.7 presents the 39 × 39 confusion matrix for FIN-Benthic2 taxonomic classification using the baseline CI loss approach, with diagonal elements (correct predictions) visually distinguished from off-diagonal elements (misclassifications) to facilitate error pattern analysis across aquatic macroinvertebrate taxa. Table A.8 provides comprehensive per-class performance metrics for all 39 aquatic macroinvertebrate taxa, detailing the taxonomic level of each taxa, true positives, support, precision, recall, and F1-scores achieved using the baseline CI loss approach, with aggregate statistics.

Table A.9 displays the 39 × 39 confusion matrix for FIN-Benthic2 taxonomic classification utilizing the Bag-CI-Aug hybrid ensemble method, enabling comparative assessment of classification improvements achieved through combined algorithmic and data-level interventions relative to baseline methodologies. Table A.10 presents detailed classification performance metrics for all 39 aquatic macroinvertebrate taxa employing the Bag-CI-Aug hybrid ensemble method, demonstrating enhanced minority class identification through integrated algorithmic and data-level interventions compared to baseline CI loss implementation. Table A.11 presents a comparative analysis of minority class performance between baseline CI loss and Bag-CI-Aug hybrid methodology, revealing heterogeneous response patterns where certain taxa demonstrate substantial recall improvements while others exhibit performance degradation, highlighting the complex interaction between ensemble strategies and taxonomic characteristics. Table A.12 organizes Bag-CI-Aug classification performance across five support-defined categories, demonstrating systematic performance enhancement correlating with increased sample availability, with small classes exhibiting substantial variance while large classes achieve consistent high performance.

Table A.13 presents comprehensive classification metrics for BIOSCAN insect identification employing baseline CI loss model across taxonomically diverse families organized by support size, revealing pronounced performance degradation in minority classes with F1-scores ranging from 0.130 to 0.845 for small classes (31–38 specimens), while large classes (5715–86,155 specimens) demonstrate consistent high performance exceeding 0.942 F1-score, achieving 94.4% overall accuracy with substantial divergence between macro (0.796) and weighted (0.944) averages indicative of class imbalance effects. Table A.14 documents enhanced classification performance utilizing Bagging with Upsampling integrated with CI loss, demonstrating systematic improvements across all support categories with small classes achieving F1-scores between 0.400–0.987 (mean improvement +14.48%), medium classes reaching 0.800–0.962 F1-scores, and large classes maintaining robust performance (0.968–0.985), resulting in 96.8% overall accuracy with improved macro-averaged metrics (0.892) indicating successful minority class performance enhancement through ensemble-based upsampling methodology.

Table A.15 lists the 12 paired Macro F1 observations used for the Wilcoxon signed-rank test on the BIOSCAN dataset, comparing the baseline CI method with the CI-US method across 3 mutually exclusive folds and 4 random seeds. Table A.16 provides a summary of the statistical significance analyses conducted for BIOSCAN and FIN-Benthic2, including the Wilcoxon signed-rank test and paired bootstrap tests used to verify the significance of the reported improvements.

Table A.5
Evaluation of hybrid ensemble strategies for the FIN-Benthic2 dataset combining data-level (up–down-sampling US-DS, augmentation) with algorithm-level (CI loss) methods through bootstrap aggregation (bagging)

Metrics	FIN-Benthic2					
	Base CI-loss	CI-Aug	CI-US-DS	Bag-CI	Bag-CI-Aug	Bag-CI-US-DS
F1-score	0.8747	0.8774	0.8751	0.8887	**0.8928**	0.8822
G-mean	0.9311	0.9354	0.9336	0.9390	**0.9402**	0.9339
BACC	0.8687	0.8762	0.8729	0.8830	**0.8850**	0.8733
MCC	0.9255	0.9407	0.9384	0.9438	**0.9526**	0.9491
Accuracy	0.9286	0.9431	0.9409	0.9460	**0.9545**	0.9511

Table A.6
Evaluation of hybrid ensemble strategies for the BIOSCAN dataset combining data-level (upsampling US, augmentation Aug) with algorithm-level (CI loss) methods through bootstrap aggregation (bagging)

Metrics	BIOSCAN					
	Base CI-loss	CI-Aug	CI-US	Bag-CI	Bag-CI-Aug	Bag-CI-US
F1-score	0.7958	0.8467	0.8684	0.8243	0.8694	**0.8922**
G-mean	0.8873	0.9157	0.9287	0.8911	0.9206	**0.9413**
BACC	0.7875	0.8387	0.8626	0.7942	0.8477	**0.8861**
MCC	0.9371	0.9571	0.9576	0.9525	0.9634	**0.9637**
Accuracy	0.9436	0.9615	0.9620	0.9575	0.9672	**0.9675**

Table A.7
Confusion matrix for aquatic macroinvertebrate classification on FIN-Benthic2 dataset using baseline CI loss model (actual vs predicted)

| Actual Class | Predicted Class | | | | |

Table A.8
Classification performance summary by class for aquatic macroinvertebrate identification using baseline CI loss model for the FIN-Benthic2 dataset.

ID	Name	Level	TP	Support	Precision	Recall	F1-score
0	*Hydropsyche saxonica*	species	80	91	0.769	0.879	0.821
1	*Nemoura cinerea*	species	38	50	0.760	0.760	0.760
2	*Capnopsis schilleri*	species	22	100	0.289	0.220	0.250
3	*Sialis* sp.	genus	187	236	0.984	0.792	0.878
4	*Leuctra nigra*	species	118	350	0.638	0.337	0.441
5	*Sphaerium* sp.	genus	349	349	1.000	1.000	1.000
6	*Silo pallipes*	species	536	538	0.95	0.996	0.973
7	*Polycentropus irroratus*	species	317	600	0.543	0.528	0.535
8	*Plectrocnemia* sp.	genus	514	704	0.775	0.729	0.752
9	*Diura* sp.	genus	893	927	0.93	0.963	0.946
10	*Protonemura* sp.	genus	751	965	0.929	0.778	0.847
11	*Ameletus inopinatus*	species	1153	1196	0.88	0.964	0.92
12	*Hydropsyche pellucidula*	species	1082	1239	0.993	0.873	0.929
13	*Rhyacophila nubila*	species	1516	1520	0.986	0.997	0.991
14	*Baetis vernus* group	species	1311	1794	0.873	0.731	0.796
15	*Nemoura* sp.	genus	1539	1850	0.882	0.832	0.856
16	*Hydraena* sp.	genus	1825	1850	0.969	0.986	0.978
17	*Oxyethira* sp.	genus	2056	2112	0.974	0.973	0.974
18	*Kageronia fuscogrisea*	species	2045	2332	0.938	0.877	0.906
19	*Lepidostoma hirtum*	species	1938	2051	0.941	0.945	0.943
20	*Polycentropus flavomaculatus*	species	1467	1900	0.825	0.772	0.797
21	*Micrasema gelidum*	species	2193	2440	0.971	0.899	0.934
22	*Isoperla* sp.	genus	2328	2600	0.756	0.895	0.82
23	*Micrasema setiferum*	species	2500	2609	0.915	0.958	0.936
24	*Agapetus* sp.	genus	2815	2881	0.961	0.977	0.969
25	*Leuctra* sp.	genus	2479	2900	0.894	0.855	0.874
26	*Ephemerella mucronata*	species	2882	3231	0.9	0.892	0.896
27	*Limnius volckmari*	species	2979	3100	0.989	0.961	0.975
28	*Amphinemura borealis*	species	2998	3300	0.932	0.908	0.92
29	*Taeniopteryx nebulosa*	species	3142	3201	0.893	0.982	0.935
30	*Ephemerella aurivillii*	species	3026	3195	0.937	0.947	0.942
31	*Oulimnius tuberculatus*	species	3205	3258	0.969	0.984	0.977
32	*Hydropsyche siltalai*	species	3854	3969	0.948	0.971	0.959
33	*Baetis rhodani*	species	3706	3989	0.929	0.866	0.896
34	*Leptophlebia* sp.	genus	3720	4045	0.979	0.920	0.948
35	*Heptagenia sulphurea*	species	3871	4234	0.925	0.914	0.92
36	*Neureclipsis bimaculata*	species	4661	4750	0.943	0.981	0.962
37	*Elmis aenea*	species	6347	6548	0.972	0.969	0.971
38	Simuliidae	family	8979	8991	0.975	0.999	0.987
Overall Accuracy		**92.9%**					
Macro Average				**0.886**	**0.869**	**0.875**	
Weighted Average				**0.929**	**0.929**	**0.928**	

Table A.9
Confusion matrix for aquatic macroinvertebrate classification on FIN-Benthic2 dataset using Bag-CI-Aug method (actual vs predicted)

Actual \ Predicted Class	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38
0	87	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	28	0	0	0	0	0	0	0	0	0	0	0	0	0	13	0	0	0	0	0	0	5	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0
2	0	0	55	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	21	0	0	24	0	0	0	0	0	0	0	0	0	0
3	0	0	0	214	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	5	0	0	16	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	37	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	307	6	0	0	0	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0	349	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6	0	0	0	0	0	0	512	0	0	0	0	0	0	0	0	0	0	0	0	26	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
7	0	0	0	0	0	0	0	120	0	0	0	0	0	0	0	0	0	0	0	0	410	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	69	0	0
8	0	0	0	9	0	0	1	22	448	0	0	0	3	0	0	19	0	0	0	7	138	0	0	0	2	0	0	12	2	6	0	0	0	0	0	0	35	0	0
9	0	0	0	0	0	0	0	0	0	885	17	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	17	0	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0	0	11	801	0	0	0	2	137	0	0	0	0	0	0	2	0	0	0	2	0	0	10	0	0	0	0	0	0	0	0	0
11	0	0	0	0	0	0	0	0	0	0	0	1183	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	6	0	0	0	0
12	5	0	0	0	0	0	0	0	0	0	0	0	1138	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	96	0	0	0	0	0	0
13	0	0	0	0	0	0	0	0	0	0	0	1	0	1519	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
14	0	0	0	0	0	0	0	0	0	0	2	0	0	0	1507	0	0	0	1	0	0	0	2	0	0	0	0	0	0	0	20	0	0	259	3	0	0	0	0
15	0	0	0	0	0	0	0	0	0	11	0	0	0	0	0	1711	0	0	0	0	0	0	27	0	0	12	28	0	29	32	0	0	0	0	0	0	0	0	0
16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1850	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
17	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2111	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2099	0	0	0	18	0	0	4	0	0	0	0	10	0	0	4	50	147	0	0	0
19	0	0	0	0	0	0	30	0	0	0	0	0	0	0	0	0	0	0	0	1966	0	44	0	11	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
20	0	0	0	0	0	0	0	52	21	0	0	52	0	0	0	1	0	0	1	4	1669	0	0	2	1	1	6	0	0	4	0	0	0	0	0	0	84	0	2
21	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	49	0	2259	0	129	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3
22	0	2	0	0	0	0	0	0	0	94	0	0	0	0	0	56	0	0	0	0	0	0	2289	1	0	61	38	0	8	17	0	0	0	4	30	0	0	0	0
23	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	45	0	33	0	2531	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2857	0	0	0	0	0	0	0	18	0	0	0	6	0	0
25	0	0	50	0	0	0	0	0	0	30	0	24	0	0	0	0	0	0	1	0	5	0	6	0	0	2761	2	1	0	2	0	0	0	0	10	0	4	0	4
26	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	11	0	0	16	3156	0	28	0	0	0	0	2	10	3	1	0	0
27	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	2985	0	0	0	14	0	0	0	0	0	93	0
28	0	0	6	0	0	0	0	0	0	0	0	0	0	0	0	56	0	4	0	0	0	0	25	0	0	4	6	0	3197	0	0	0	0	0	0	0	2	0	
29	0	0	0	0	0	0	0	0	0	0	5	0	0	0	0	13	0	0	0	0	0	0	0	0	0	8	0	0	0	3175	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	107	0	0	0	3086	0	0	0	1	1	0	0	0
31	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	2	0	0	3222	0	0	0	0	0	33	0	
32	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	0	0	0	4	0	0	0	4	0	0	3950	0	0	0	0	4	0	0
33	0	0	0	0	0	0	0	0	0	0	0	0	0	0	210	0	0	0	0	0	0	0	0	0	0	0	12	0	14	0	0	0	0	3725	27	0	0	0	1
34	0	0	0	0	0	0	0	0	1	0	0	4	0	0	0	0	0	1	13	1	1	0	0	0	0	20	22	0	9	0	5	0	0	2	3947	11	1	0	7
35	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	12	12	0	0	0	0	0	0	0	0	0	0	0	10	0	0	0	10	4177	0	0	11
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	0	0	0	4	0	0	0	1	0	0	5	0	0	0	0	4730	0	1
37	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	6	0	0	18	0	0	0	0	0	6523	0
38	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	19	0	18	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	8948

Table A.10
Classification performance summary by class for aquatic macroinvertebrate identification using Bag-CI-Aug method for the FIN-Benthic2 dataset.

ID	Name	Level	TP	Support	Precision	Recall	F1-score
0	*Hydropsyche saxonica*	species	87	91	0.946	0.956	0.951
1	*Nemoura cinerea*	species	28	50	0.933	0.560	0.700
2	*Capnopsis schilleri*	species	55	100	0.482	0.550	0.514
3	*Sialis* sp.	genus	214	236	0.960	0.907	0.932
4	*Leuctra nigra*	species	37	350	1.000	0.106	0.191
5	*Sphaerium* sp.	genus	349	349	1.000	1.000	1.000
6	*Silo pallipes*	species	512	538	0.943	0.952	0.947
7	*Polycentropus irroratus*	species	120	600	0.619	0.200	0.302
8	*Plectrocnemia* sp.	genus	448	704	0.951	0.636	0.763
9	*Diura* sp.	genus	885	927	0.868	0.955	0.909
10	*Protonemura* sp.	genus	801	965	0.958	0.83	0.89
11	*Ameletus inopinatus*	species	1183	1196	0.935	0.989	0.961
12	*Hydropsyche pellucidula*	species	1138	1239	0.993	0.918	0.954
13	*Rhyacophila nubila*	species	1519	1520	1.0	0.999	1.0
14	*Baetis vernus* group	species	1507	1794	0.877	0.84	0.858
15	*Nemoura* sp.	genus	1711	1850	0.85	0.925	0.886
16	*Hydraena* sp.	genus	1850	1850	1.0	1.0	1.0
17	*Oxyethira* sp.	genus	2111	2112	0.986	1.0	0.993

(*continued on next page*)

Table A.10 (*continued*).

ID	Name	Level	TP	Support	Precision	Recall	F1-score
18	*Kageronia fuscogrisea*	species	2099	2332	0.987	0.9	0.941
19	*Lepidostoma hirtum*	species	1966	2051	0.929	0.959	0.943
20	*Polycentropus flavomaculatus*	species	1669	1900	0.745	0.878	0.806
21	*Micrasema gelidum*	species	2259	2440	0.96	0.926	0.942
22	*Isoperla* sp.	genus	2289	2600	0.958	0.88	0.917
23	*Micrasema setiferum*	species	2531	2609	0.947	0.97	0.958
24	*Agapetus* sp.	genus	2857	2881	0.996	0.992	0.994
25	*Leuctra* sp.	genus	2761	2900	0.855	0.952	0.901
26	*Ephemerella mucronata*	species	3156	3231	0.931	0.977	0.953
27	*Limnius volckmari*	species	2985	3100	0.996	0.963	0.979
28	*Amphinemura borealis*	species	3197	3300	0.961	0.969	0.965
29	*Taeniopteryx nebulosa*	species	3175	3201	0.973	0.992	0.982
30	*Ephemerella aurivillii*	species	3086	3195	0.986	0.966	0.976
31	*Oulimnius tuberculatus*	species	3222	3258	0.99	0.989	0.99
32	*Hydropsyche siltalai*	species	3950	3969	0.971	0.995	0.983
33	*Baetis rhodani*	species	3725	3989	0.931	0.934	0.932
34	*Leptophlebia* sp.	genus	3947	4045	0.964	0.976	0.97
35	*Heptagenia sulphurea*	species	4177	4234	0.963	0.987	0.974
36	*Neureclipsis bimaculata*	species	4730	4750	0.959	0.996	0.977
37	*Elmis aenea*	species	6523	6548	0.981	0.996	0.988
38	Simuliidae	family	8948	8991	0.997	0.995	0.996
Overall Accuracy		**95.4%**					
Macro Average				**0.930**	**0.885**	**0.893**	
Weighted Average				**0.955**	**0.954**	**0.952**	

Table A.11
Performance comparison: CI loss vs Bag-CI-Aug for key minority classes for the FIN-Benthic2 dataset.

Class	CI loss		Bag-CI-Aug		Change	Impact
	TP/Total	Recall	TP/Total	Recall		
N. cinerea	38/50	76.0%	28/50	56.0%	−20.0%	Decline
H. saxonica	80/91	87.9%	87/91	95.6%	+7.7%	Minor Improvement
C. schilleri	22/100	22.0%	55/100	55.0%	+33.0%	Major Improvement
Sialis sp.	187/236	79.2%	214/236	90.7%	+11.5%	Good Improvement
Sphaerium sp.	349/349	100.0%	349/349	100.0%	+0.0%	Stable
L. nigra	118/350	33.7%	37/350	10.6%	−23.1%	Decline

Table A.12
Class imbalance analysis: Bag-CI-Aug method performance by support size for the FIN-Benthic2 dataset.

Support range	Classes	Avg precision	Avg recall	Avg F1	Best/worst
Very Small (50–100)	3	0.787	0.689	0.722	Best: Class 0 (0.951) Worst: Class 2 (0.514)
Small (101–500)	3	0.987	0.671	0.708	Best: Class 5 (1.000) Worst: Class 4 (0.191)
Medium (501–1000)	5	0.868	0.715	0.762	Best: Class 6 (0.947) Worst: Class 7 (0.302)
Large (1001–3000)	15	0.934	0.942	0.937	Best: Class 16 (1.000) Worst: Class 20 (0.806)
Very Large (3001+)	13	0.969	0.980	0.974	Best: Class 38 (0.996) Worst: Class 33 (0.932)

Key Insights
•Performance improves significantly for classes with >1000 support size (F1 >93%)
•Small classes (101–500) exhibit highest performance variance (range: 0.191–1.000)
•Performance stability increases substantially for larger support sizes
•Very Large classes (3001+) achieve most consistent results (F1 variance: 0.064)

Table A.13
Classification performance summary by class for BIOSCAN insect identification using baseline CI loss model.

ID	Name	TP	Support	Precision	Recall	F1-score
Small Classes (31–38 specimens)						
0	*Opostegidae*	23	31	0.742	0.742	0.742
1	*Agaonidae*	23	32	0.767	0.719	0.742
2	*Laemophloeidae*	28	33	0.757	0.848	0.800
3	*Biphyllidae*	26	33	0.591	0.788	0.675
4	*Mycteridae*	24	34	0.857	0.706	0.774
5	*Coleophoridae*	3	35	0.273	0.086	0.130
6	*Peripsocidae*	28	35	0.875	0.800	0.836
7	*Tingidae*	27	36	0.900	0.750	0.818
8	*Archipsocidae*	30	37	0.882	0.811	0.845
9	*Chrysididae*	27	38	0.931	0.711	0.806
Medium Classes (102–131 specimens)						
10	*Cantharidae*	84	102	0.700	0.824	0.757
11	*Endomychidae*	98	106	0.942	0.925	0.933
12	*Phengodidae*	105	109	0.833	0.963	0.894
13	*Ditomyiidae*	75	114	0.641	0.658	0.649
14	*Eucnemidae*	116	122	0.829	0.951	0.885
15	*Phalacridae*	109	127	0.838	0.858	0.848
16	*Trigonidiidae*	77	127	0.837	0.606	0.703
17	*Fanniidae*	76	129	0.598	0.589	0.594
18	*Torymidae*	102	131	0.703	0.779	0.739
19	*Liposcelididae*	121	131	0.938	0.924	0.931
Large Classes (5715–86155 specimens)						
20	*Scelionidae*	5446	5715	0.931	0.953	0.942
21	*Sphaeroceridae*	5726	5929	0.969	0.966	0.967
22	*Cicadellidae*	5764	5951	0.951	0.969	0.960
23	*Dolichopodidae*	11 474	11 886	0.955	0.965	0.960
24	*Psychodidae*	14 802	15 267	0.932	0.970	0.950
25	*Ceratopogonidae*	25 373	26 280	0.968	0.965	0.967
26	*Sciaridae*	29 649	30 843	0.952	0.961	0.957
27	*Phoridae*	44 537	45 414	0.970	0.981	0.975
28	*Chironomidae*	47 710	50 180	0.972	0.951	0.961
29	*Cecidomyiidae*	82 602	86 155	0.977	0.959	0.968
Overall Accuracy		**94.4%**				
Macro Average				**0.819**	**0.787**	**0.796**
Weighted Average				**0.945**	**0.944**	**0.944**

Table A.14
Classification performance summary by class for BIOSCAN insect identification using bagging with upsampling and CI loss.

ID	Name	TP	Support	Precision	Recall	F1-score
Small Classes (31–38 specimens)						
0	*Opostegidae*	28	31	0.966	0.903	0.933
1	*Agaonidae*	29	32	0.853	0.906	0.879
2	*Laemophloeidae*	31	33	0.912	0.939	0.925
3	*Biphyllidae*	33	33	0.943	1.000	0.971
4	*Mycteridae*	30	34	0.909	0.882	0.896
5	*Coleophoridae*	11	35	0.550	0.314	0.400
6	*Peripsocidae*	26	35	0.867	0.743	0.800
7	*Tingidae*	35	36	0.972	0.972	0.972
8	*Archipsocidae*	37	37	0.974	1.000	0.987
9	*Chrysididae*	33	38	1.000	0.868	0.930
Medium Classes (102–131 specimens)						
10	*Cantharidae*	85	102	0.944	0.833	0.885
11	*Endomychidae*	101	106	0.962	0.953	0.957
12	*Phengodidae*	105	109	0.897	0.963	0.929
13	*Ditomyiidae*	95	114	0.785	0.833	0.809
14	*Eucnemidae*	118	122	0.894	0.967	0.929
15	*Phalacridae*	126	127	0.933	0.992	0.962
16	*Trigonidiidae*	119	127	0.838	0.937	0.885
17	*Fanniidae*	102	129	0.810	0.791	0.800
18	*Torymidae*	110	131	0.948	0.840	0.891
19	*Liposcelididae*	127	131	0.907	0.969	0.937

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Table A.14 (*continued*).

ID	Name	TP	Support	Precision	Recall	F1-score
Large Classes (5715–86155 specimens)						
20	*Scelionidae*	5554	5715	0.965	0.972	0.968
21	*Sphaeroceridae*	5830	5929	0.982	0.983	0.983
22	*Cicadellidae*	5865	5951	0.985	0.986	0.985
23	*Dolichopodidae*	11 678	11 886	0.966	0.983	0.974
24	*Psychodidae*	14 882	15 267	0.969	0.975	0.972
25	*Ceratopogonidae*	25 747	26 280	0.984	0.980	0.982
26	*Sciaridae*	30 006	30 843	0.974	0.973	0.973
27	*Phoridae*	44 916	45 414	0.975	0.989	0.982
28	*Chironomidae*	48 503	50 180	0.983	0.967	0.975
29	*Cecidomyiidae*	84 373	86 155	0.979	0.979	0.979
Overall Accuracy		**96.8%**				
Macro Average				**0.904**	**0.886**	**0.892**
Weighted Average				**0.968**	**0.968**	**0.967**

Table A.15
Paired Macro F1-scores used for the Wilcoxon signed-rank test on the BIOSCAN dataset (CI vs. CI-US)

Fold-Seed	CI	CI-US	Difference (CI-US - CI)
f0_42	0.8045	0.8695	0.0650
f0_123	0.8063	0.8675	0.0612
f0_999	0.8048	0.8715	0.0667
f0_2024	0.8015	0.8721	0.0706
f1_42	0.8078	0.8707	0.0629
f1_123	0.8038	0.8714	0.0676
f1_999	0.8122	0.8739	0.0617
f1_2024	0.7985	0.8714	0.0729
f2_42	0.8121	0.8701	0.0580
f2_123	0.8062	0.8713	0.0651
f2_999	0.8112	0.8735	0.0623
f2_2024	0.8115	0.8708	0.0593
Mean	**0.8067**	**0.8711**	**0.0644**

Table A.16
Statistical significance analysis for the BIOSCAN and FIN-Benthic2 datasets.

Dataset	Comparison/metric	Method	p-value	Interpretation
FIN-Benthic2	Bag-CI-Aug vs. CI (Accuracy)	Paired bootstrap test	<0.0001	Statistically significant improvement
FIN-Benthic2	Bag-CI-Aug vs. CI (Macro F1)	Paired bootstrap test	<0.0001	Statistically significant improvement
BIOSCAN	CI vs. CI-US (Macro F1, 12 paired runs)	Wilcoxon signed-rank test	0.000488	Statistically significant improvement of CI-US over CI
BIOSCAN	Bag-CI-US vs. CI (Accuracy)	Paired bootstrap test	<0.0001	Statistically significant improvement
BIOSCAN	Bag-CI-US vs. CI (Macro F1)	Paired bootstrap test	<0.0001	Statistically significant improvement

Data availability

The FIN-Benthic2 and BIOSCAN-1M datasets are available through their respective repositories at https://etsin.fairdata.fi/dataset/8e41270d-4580-4766-9415-21b549443b23 and https://github.com/bioscan-ml/BIOSCAN-1M. All original code used for implementation and comparative analysis is publicly available on GitHub at https://github.com/mskhan793/taxonomist/tree/imbalance.

References

Abdani, S. R., Zulkifley, M. A., & Zulkifley, N. H. (2022). Undersampling and oversampling strategies for convolutional neural networks classifier. In *Lecture notes in electrical engineering*: *vol. 842*, *Proceedings of the 6th international conference on electrical, control and computer engineering* (pp. 1129–1137). Singapore: Springer, http://dx.doi.org/10.1007/978-981-16-8690-0_98.

Abdelhalim, I. S. A., Mohamed, M. F., & Mahdy, Y. B. (2021). Data augmentation for skin lesion using self-attention based progressive generative adversarial network. *Expert Systems with Applications*, *165*, Article 113922. http://dx.doi.org/10.1016/j.eswa.2020.113922.

Araf, I., Idri, A., & Chairi, I. (2024). Cost-sensitive learning for imbalanced medical data: A review. *Artificial Intelligence Review*, *57*(4), 80. http://dx.doi.org/10.1007/s10462-023-10652-8.

Ärje, J., Raitoharju, J., Iosifidis, A., Tirronen, V., Meissner, K., Gabbouj, M., Kiranyaz, S., & Kärkkäinen, S. (2020). Human experts vs. machines in taxa recognition. *Signal Processing: Image Communication*, *87*, Article 115917. http://dx.doi.org/10.1016/j.image.2020.115917.

Arjovsky, M., Chintala, S., & Bottou, L. (2017). Wasserstein generative adversarial networks. In D. Precup, & Y. W. Teh (Eds.), *Proceedings of machine learning research*: *vol. 70*, *Proceedings of the 34th international conference on machine learning* (pp. 214–223). PMLR, URL: https://proceedings.mlr.press/v70/arjovsky17a.html.

Arun Pandian, J., Geetharamani, G., & Annette, B. (2019). Data augmentation on plant leaf disease image dataset using image manipulation and deep learning techniques. In *2019 IEEE 9th international conference on advanced computing* (pp. 199–204). http://dx.doi.org/10.1109/IACC48062.2019.8971580.

Belmont, J., Miller, C., Scott, M., & Wilkie, C. (2022). A new statistical approach for identifying rare species under imperfect detection. *Diversity and Distributions*, *28*(5), 882–893. http://dx.doi.org/10.1111/ddi.13495.

Benkendorf, D. J., Schwartz, S. D., Cutler, D. R., & Hawkins, C. P. (2023). Correcting for the effects of class imbalance improves the performance of machine-learning based species distribution models. *Ecological Modelling*, *483*, Article 110414. http://dx.doi.org/10.1016/j.ecolmodel.2023.110414.

Bhatia, S., & Dahyot, R. (2019). Using WGAN for improving imbalanced classification performance. *vol. 2563*, In *CEUR workshop proceedings* (pp. 365–375). URL: https://mural.maynoothuniversity.ie/id/eprint/15250/.

Binta Islam, S., Valles, D., Hibbitts, T. J., Ryberg, W. A., Walkup, D. K., & Forstner, M. R. J. (2023). Animal species recognition with deep convolutional neural networks from ecological camera trap images. *Animals*, *13*(9), 1526. http://dx.doi.org/10.3390/ani13091526.

Bjerge, K., Alison, J., Dyrmann, M., Frigaard, C. E., Mann, H. M., & Høye, T. T. (2023). Accurate detection and identification of insects from camera trap images with deep learning. *PLOS Sustainability and Transformation*, *2*(3), Article e0000051. http://dx.doi.org/10.1371/journal.pstr.0000051.

Buda, M., Maki, A., & Mazurowski, M. A. (2018). A systematic study of the class imbalance problem in convolutional neural networks. *Neural Networks*, *106*, 249–259. http://dx.doi.org/10.1016/j.neunet.2018.07.011.

Carvalho, M., Pinho, A. J., & Brás, S. (2025). Resampling approaches to handle class imbalance: a review from a data perspective. *Journal of Big Data*, *12*(1), 71. http://dx.doi.org/10.1186/s40537-025-01119-4.

Chen, S., Gao, Q., & He, Y. (2024). A pest image recognition method for long-tail distribution problem. *Frontiers in Environmental Science*, *12*, Article 1391770. http://dx.doi.org/10.3389/fenvs.2024.1391770.

Chen, C., Liaw, A., & Breiman, L. (2004). Using random forest to learn imbalanced data. *University of California, Berkeley*, *110*(1–12), 24.

Chen, W., Yang, K., Yu, Z., Shi, Y., & Chen, C. P. (2024). A survey on imbalanced learning: Latest research, applications and future directions. *Artificial Intelligence Review*, *57*(6), 137. http://dx.doi.org/10.1007/s10462-024-10759-6.

Cordier, T., Lanzén, A., Apothéloz-Perret-Gentil, L., Stoeck, T., & Pawlowski, J. (2019). Embracing environmental genomics and machine learning for routine biomonitoring. *Trends in Microbiology*, *27*(5), 387–397. http://dx.doi.org/10.1016/j.tim.2018.10.012.

Cruz, L. D., Lopez, D. M., Vargas-Canas, R., Figueroa, A., & Corrales, J. C. (2022). Computer-assisted bioidentification using freshwater macroinvertebrates: A scoping review. *Water*, *14*(20), Article 3249. http://dx.doi.org/10.3390/w14203249.

de la Cruz Huayanay, A., Bazán, J. L., & Russo, C. M. (2025). Performance of evaluation metrics for classification in imbalanced data. *Computational Statistics*, *40*, 1447–1473. http://dx.doi.org/10.1007/s00180-024-01539-5.

Cui, J., Zhong, Z., Liu, S., Yu, B., & Jia, J. (2021). Parametric contrastive learning. In *2021 IEEE/CVF international conference on computer vision* (pp. 695–704). http://dx.doi.org/10.1109/ICCV48922.2021.00075.

de Pedro Noriega, L., Bobo-Pinilla, J., Delgado-Iglesias, J., Reinoso-Tapia, R., Gallego, A. M., & Quirós-Alpera, S. (2025). AI in biodiversity education: The bias in endangered species information and its implications. *Sustainability*, *17*(14), 6554. http://dx.doi.org/10.3390/su17146554.

De Souza, V. L. T., Marques, B. A. D., Batagelo, H. C., & Gois, J. P. (2023). A review on generative adversarial networks for image generation. *Computers & Graphics*.

Duarte, D., Andriolo, U., & Gonçalves, G. (2020). Addressing the class imbalance problem in the automatic image classification of coastal litter from orthophotos derived from UAS imagery. *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, *vol. V-3-2020*, 439–445. http://dx.doi.org/10.5194/isprs-annals-V-3-2020-439-2020.

Farhadpour, S., Warner, T. A., & Maxwell, A. E. (2024). Selecting and interpreting multiclass loss and accuracy assessment metrics for classifications with class imbalance: Guidance and best practices. *Remote Sensing*, *16*(3), 533. http://dx.doi.org/10.3390/rs16030533.

Francazi, E., Baity-Jesi, M., & Lucchi, A. (2023). A theoretical analysis of the learning dynamics under class imbalance. In A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, & J. Scarlett (Eds.), *Proceedings of machine learning research*: *vol. 202*, *Proceedings of the 40th international conference on machine learning* (pp. 10285–10322). PMLR, URL: https://proceedings.mlr.press/v202/francazi23a.html.

Galar, M., Fernandez, A., Barrenechea, E., Bustince, H., & Herrera, F. (2012). A review on ensembles for the class imbalance problem: Bagging-, boosting-, and hybrid-based approaches. *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)*, *42*(4), 463–484. http://dx.doi.org/10.1109/TSMCC.2011.2161285.

Gharaee, Z., Gong, Z., Pellegrino, N., Zarubiieva, I., Haurum, J. B., Lowe, S., McKeown, J., Ho, C., McLeod, J., Wei, Y.-Y., Agda, J., Ratnasingham, S., Steinke, D., Chang, A., Taylor, G. W., & Fieguth, P. (2023). A step towards worldwide biodiversity assessment: The BIOSCAN-1m insect dataset. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, & S. Levine (Eds.), *36*, *Advances in neural information processing systems* (pp. 43593–43619). Curran Associates, Inc., URL: https://proceedings.neurips.cc/paper_files/paper/2023/file/87dbbdc3a685a97ad28489a1d57c45c1-Paper-Datasets_and_Benchmarks.pdf.

Ghosh, A., Sufian, A., Sultana, F., Chakrabarti, A., & De, D. (2020). Fundamental concepts of convolutional neural network. In *Intelligent systems reference library*: *vol. 172*, *Recent trends and advances in artificial intelligence and internet of things* (pp. 519–567). Cham: Springer, http://dx.doi.org/10.1007/978-3-030-32644-9_36.

Goldsworthy, C., & Matam, B. R. (2021). Improved rare species identification using focal loss based deep learning models. *International Journal of Computer and Information Engineering*, *15*(1), 18–22.

Goodfellow, I. J., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., & Bengio, Y. (2014). Generative adversarial nets. In Z. Ghahramani, M. Welling, C. Cortes, N. Lawrence, & K. Weinberger (Eds.), *27*, *Advances in neural information processing systems*. Curran Associates, Inc., URL: https://proceedings.neurips.cc/paper_files/paper/2014/file/f033ed80deb0234979a61f95710dbe25-Paper.pdf.

van den Goorbergh, R., van Smeden, M., Timmerman, D., & Van Calster, B. (2022). The harm of class imbalance corrections for risk prediction models: illustration and simulation using logistic regression. *Journal of the American Medical Informatics Association*, *29*(9), 1525–1534. http://dx.doi.org/10.1093/jamia/ocac093.

Gorodkin, J. (2004). Comparing two K-category assignments by a K-category correlation coefficient. *Computational Biology and Chemistry*, *28*(5), 367–374. http://dx.doi.org/10.1016/j.compbiolchem.2004.09.006.

He, C., Shen, Y., Fang, C., Xiao, F., Tang, L., Zhang, Y., Zuo, W., Guo, Z., & Li, X. (2025). Diffusion models in low-level vision: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *47*(6), 4630–4651. http://dx.doi.org/10.1109/TPAMI.2025.3545047.

Howard, A. G., Zhu, M., Chen, B., Kalenichenko, D., Wang, W., Weyand, T., Andreetto, M., & Adam, H. (2017). MobileNets: Efficient convolutional neural networks for mobile vision applications. http://dx.doi.org/10.48550/arXiv.1704.04861, arXiv preprint arXiv:1704.04861.

Høye, T. T., Ärje, J., Bjerge, K., Hansen, O. L. P., Iosifidis, A., Leese, F., Mann, H. M. R., Meissner, K., Melvad, C., & Raitoharju, J. (2021). Deep learning and computer vision will transform entomology. *Proceedings of the National Academy of Sciences*, *118*(2), Article e2002545117. http://dx.doi.org/10.1073/pnas.2002545117.

Høye, T. T., Dyrmann, M., Kjær, C., Nielsen, J., Bruus, M., Mielec, C. L., Vesterdal, M. S., Bjerge, K., Madsen, S. A., Jeppesen, M. R., & Melvad, C. (2022). Accurate image-based identification of macroinvertebrate specimens using deep learning—How much training data is needed? *PeerJ*, *10*, Article e13837. http://dx.doi.org/10.7717/peerj.13837.

Huang, C. Y., & Dai, H. L. (2021). Learning from class-imbalanced data: review of data driven methods and algorithm driven methods. *Data Science in Finance and Economics*, *1*(1), 21–36. http://dx.doi.org/10.3934/DSFE.2021002.

Huang, Y., Fields, K. G., & Ma, Y. (2022). A tutorial on generative adversarial networks with application to classification of imbalanced data. *Statistical Analysis and Data Mining: The ASA Data Science Journal*, *15*(5), 543–552. http://dx.doi.org/10.1002/sam.11570.

Hung, S.-K., & Gan, J. Q. (2021). Augmentation of small training data using GANs for enhancing the performance of image classification. In *2020 25th international conference on pattern recognition (ICPR)* (pp. 3350–3356). IEEE, http://dx.doi.org/10.1109/ICPR48806.2021.9412399.

Islam, M. R., Hassan, R., Nazib, A., Nguyen, K., Fookes, C., & Islam, M. Z. (2024). Size and smoothness aware adaptive focal loss for small tumor segmentation. http://dx.doi.org/10.48550/arXiv.2407.09828, arXiv preprint arXiv:2407.09828.

Jafarigol, E., Trafalis, T., & Mohammadi, N. (2023). A review of machine learning techniques in imbalanced data and future trends. http://dx.doi.org/10.48550/arXiv.2310.07917, arXiv preprint arXiv:2310.07917.

Johnson, J. M., & Khoshgoftaar, T. M. (2019). Survey on deep learning with class imbalance. *Journal of Big Data*, *6*(1), Article 27. http://dx.doi.org/10.1186/s40537-019-0192-5.

Joloudari, J. H., Marefat, A., Nematollahi, M. A., Oyelere, S. S., & Hussain, S. (2023). Effective class-imbalance learning based on SMOTE and convolutional neural networks. *Applied Sciences*, *13*(6), 4006. http://dx.doi.org/10.3390/app13064006.

Kang, S.-H., Jeong, I.-S., & Lim, H.-S. (2024). A deep learning-based biomonitoring system for detecting water pollution using caenorhabditis elegans swimming behaviors. *Ecological Informatics*, *80*, Article 102482. http://dx.doi.org/10.1016/j.ecoinf.2024.102482.

Karras, T., Aila, T., Laine, S., & Lehtinen, J. (2018). Progressive growing of GANs for improved quality, stability, and variation. In *International conference on learning representations*.

Karras, T., Aittala, M., Laine, S., Härkönen, E., Hellsten, J., Lehtinen, J., & Aila, T. (2021). Alias-free generative adversarial networks. *vol. 34*, In *Advances in neural information processing systems* (pp. 852–863). Curran Associates, Inc., URL: https://proceedings.neurips.cc/paper_files/paper/2021/file/076ccd93ad68be51f23707988e934906-Paper.pdf.

Karras, T., Laine, S., & Aila, T. (2019). A style-based generator architecture for generative adversarial networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 4401–4410).

Karras, T., Laine, S., Aittala, M., Hellsten, J., Lehtinen, J., & Aila, T. (2020). Analyzing and improving the image quality of StyleGAN. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 8110–8119).

Keck, F., Vasselon, V., Tapolczai, K., Rimet, F., & Bouchez, A. (2017). Freshwater biomonitoring in the information age. *Frontiers in Ecology and the Environment*, *15*(5), 266–274. http://dx.doi.org/10.1002/fee.1490.

Khan, A. A., Chaudhari, O., & Chandra, R. (2024). A review of ensemble learning and data augmentation models for class imbalanced problems: Combination, implementation and evaluation. *Expert Systems with Applications*, *244*, Article 122778. http://dx.doi.org/10.1016/j.eswa.2023.122778.

Kim, M., & Hwang, K.-B. (2022). An empirical evaluation of sampling methods for the classification of imbalanced data. *PLoS ONE*, *17*(7), Article e0271260. http://dx.doi.org/10.1371/journal.pone.0271260.

Klasen, M., Ahrens, D., Eberle, J., & Steinhage, V. (2022). Image-based automated species identification: Can virtual data augmentation overcome problems of insufficient sampling? *Systematic Biology*, *71*(2), 320–333. http://dx.doi.org/10.1093/sysbio/syab048.

Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). ImageNet classification with deep convolutional neural networks. *vol. 25*, In *Advances in neural information processing systems* (pp. 1097–1105). Curran Associates, Inc., URL: https://proceedings.neurips.cc/paper_files/paper/2012/file/c399862d3b9d6b76c8436e924a68c45b-Paper.pdf.

Kupas, D., & Harangi, B. (2021). Solving the problem of imbalanced dataset with synthetic image generation for cell classification using deep learning. In *2021 43rd annual international conference of the IEEE engineering in medicine & biology society* (pp. 2981–2984). IEEE, http://dx.doi.org/10.1109/EMBC46164.2021.9631065.

Kwon, Y. S., Kang, H., & Pyo, J. (2024). Estimation of aquatic ecosystem health using deep neural network with nonlinear data mapping. *Ecological Informatics*, *81*, Article 102588. http://dx.doi.org/10.1016/j.ecoinf.2024.102588.

Lee, Y. S., & Bang, C. C. (2022). Framework for the classification of imbalanced structured data using under-sampling and convolutional neural network. *Information Systems Frontiers*, *24*(6), 1795–1809. http://dx.doi.org/10.1007/s10796-021-10195-9.

Lemaître, G., Nogueira, F., & Aridas, C. K. (2017). Imbalanced-learn: A Python toolbox to tackle the curse of imbalanced datasets in machine learning. *Journal of Machine Learning Research*, *18*(17), 1–5, URL: http://jmlr.org/papers/v18/16-365.html.

Li, Z., Liu, F., Yang, W., Peng, S., & Zhou, J. (2022). A survey of convolutional neural networks: Analysis, applications, and prospects. *IEEE Transactions on Neural Networks and Learning Systems*, *33*(12), 6999–7019. http://dx.doi.org/10.1109/TNNLS.2021.3084827.

Lim, J., Han, D., Nejad, M. P. S., & Chon, K. H. (2024). ECG classification via integration of adaptive beat segmentation and relative heart rate with deep learning networks. *Computers in Biology and Medicine*, *181*, Article 109062. http://dx.doi.org/10.1016/j.compbiomed.2024.109062.

Lin, T.-Y., Goyal, P., Girshick, R., He, K., & Dollár, P. (2017). Focal loss for dense object detection. In *Proceedings of the IEEE international conference on computer vision* (pp. 2980–2988).

Ling, C. X., & Sheng, V. S. (2008). Cost-sensitive learning and the class imbalance problem. In C. Sammut (Ed.), *Encyclopedia of machine learning* (pp. 231–235). Springer.

Liu, S., Wang, Y., Zhang, J., Chen, C., & Xiang, Y. (2017). Addressing the class imbalance problem in Twitter spam detection using ensemble learning. *Computers & Security*, *69*, 35–49. http://dx.doi.org/10.1016/j.cose.2016.12.004.

Loshchilov, I., & Hutter, F. (2017). Decoupled weight decay regularization. http://dx.doi.org/10.48550/arXiv.1711.05101, arXiv preprint arXiv:1711.05101.

MacIsaac, J., Newson, S., Ashton-Butt, A., Pearce, H., & Milner, B. (2024). Improving acoustic species identification using data augmentation within a deep learning framework. *Ecological Informatics*, *83*, Article 102851. http://dx.doi.org/10.1016/j.ecoinf.2024.102851.

Martinez, B., Reaser, J. K., Dehgan, A., et al. (2020). Technology innovation: Advancing capacities for the early detection of and rapid response to invasive species. *Biological Invasions*, *22*, 75–100. http://dx.doi.org/10.1007/s10530-019-02146-y.

Miller, T., Michoński, G., Durlik, I., Kozlovska, P., & Biczak, P. (2025). Artificial intelligence in aquatic biodiversity research: A PRISMA-based systematic review. *Biology*, *14*(5), 520. http://dx.doi.org/10.3390/biology14050520.

Miloˇsević, D., Milosavljević, A., Predić, B., Medeiros, A. S., Savić-Zdravković, D., Stojković Piperac, M., Kostić, T., Spasić, F., & Leese, F. (2020). Application of deep learning in aquatic bioassessment: Towards automated identification of non-biting midges. *Science of the Total Environment*, *711*, Article 135160. http://dx.doi.org/10.1016/j.scitotenv.2019.135160.

Mohammadisabet, A., Hasan, R., Dattana, V., Mahmood, S., & Hussain, S. (2025). CNN-based optimization for fish species classification: Tackling environmental variability, class imbalance, and real-time constraints. *Information*, *16*(2), 154. http://dx.doi.org/10.3390/info16020154.

Moorthy, R. S., Selvaraj, A. K., Prathiba, S. B., Yenduri, G., Mohanty, S. N., & Ramesh, J. V. N. (2024). Handling the class imbalance problem with an improved sine cosine algorithm for optimal instance selection. *IEEE Access*, *12*, 87131–87151. http://dx.doi.org/10.1109/ACCESS.2024.3417822.

Nazari, E., & Branco, P. (2021). On oversampling via generative adversarial networks under different data difficulty factors. In N. Moniz, P. Branco, L. Torgo, N. Japkowicz, M. Woźniak, & S. Wang (Eds.), *Proceedings of machine learning research*: *vol. 154*, *Proceedings of the third international workshop on learning with imbalanced domains: theory and applications* (pp. 76–89). PMLR, URL: https://proceedings.mlr.press/v154/nazari21a.html.

Owusu-Adjei, M., Hayfron-Acquah, J. B., Frimpong, T., & Abdul-Salaam, G. (2023). Imbalanced class distribution and performance evaluation metrics: A systematic review of prediction accuracy for determining model performance in healthcare systems. *PLOS Digital Health*, *2*(11), Article e0000290. http://dx.doi.org/10.1371/journal.pdig.0000290.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, *12*, 2825–2830, URL: https://www.jmlr.org/papers/volume12/pedregosa11a/pedregosa11a.pdf.

Radford, A., Metz, L., & Chintala, S. (2016). Unsupervised representation learning with deep convolutional generative adversarial networks. In *International conference on learning representations*. http://dx.doi.org/10.48550/arXiv.1511.06434.

Rahimi Mashkale, M., Delavar, M. A., & Jamshidi, M. (2025). Using of cost sensitive learning approach for prediction of imbalanced soil classes. *Journal of Soil Management and Sustainable Production*, *14*(4), 53–73.

Rahman, Z. U., Asaari, M. S. M., Ibrahim, H., Abidin, I. S. Z., & Ishak, M. K. (2024). Generative adversarial networks (GANs) for image augmentation in farming: A review. *IEEE Access*, *12*, 179912–179943. http://dx.doi.org/10.1109/ACCESS.2024.3505989.

Raitoharju, J. (2022). Convolutional neural networks. In A. Iosifidis, & A. Tefas (Eds.), *Deep learning for robot perception and cognition* (pp. 35–69). Academic Press, http://dx.doi.org/10.1016/B978-0-323-85787-1.00008-7.

Raitoharju, J., Ärje, J., Iosifidis, A., Tirronen, V., Meissner, K., Gabbouj, M., Kiranyaz, S., Kärkkäinen, S., et al. (2018). FIN-benthic 2. Finnish Environment Institute, http://urn.fi/urn:nbn:fi:csc-kata20181023162421949328.

Raitoharju, J., & Meissner, K. (2019). On confidences and their use in (semi-)automatic multi-image taxa identification. In *2019 IEEE symposium series on computational intelligence* (pp. 1338–1343). IEEE, http://dx.doi.org/10.1109/SSCI44817.2019.9002975.

Raitoharju, J., Riabchenko, E., Ahmad, I., Iosifidis, A., Gabbouj, M., Kiranyaz, S., Tirronen, V., Ärje, J., Kärkkäinen, S., & Meissner, K. (2018). Benchmark database for fine-grained image classification of benthic macroinvertebrates. *Image and Vision Computing*, *78*, 73–83. http://dx.doi.org/10.1016/j.imavis.2018.06.005.

Resh, V. H., & Rosenberg, D. M. (1993). *Freshwater biomonitoring and benthic macroinvertebrates*. Chapman & Hall.

Sagova-Mareckova, M., Boenigk, J., Bouchez, A., Cermakova, K., Chonova, T., Cordier, T., Eisendle, U., Elersek, T., Fazi, S., Fleituch, T., & Stoeck, T. (2021). Expanding ecological assessment by integrating microorganisms into routine freshwater biomonitoring. *Water Research*, *191*, Article 116767. http://dx.doi.org/10.1016/j.watres.2020.116767.

Saini, M., & Susan, S. (2020). Deep transfer with minority data augmentation for imbalanced breast cancer dataset. *Applied Soft Computing*, *97*, Article 106759. http://dx.doi.org/10.1016/j.asoc.2020.106759.

Saini, M., & Susan, S. (2023). Tackling class imbalance in computer vision: A contemporary review. *Artificial Intelligence Review*, *56*(1), 1279–1335. http://dx.doi.org/10.1007/s10462-023-10557-6.

Sampath, V., Maurtua, I., Aguilar Martín, J. J., & Gutierrez, A. (2021). A survey on generative adversarial networks for imbalance problems in computer vision tasks. *Journal of Big Data*, *8*(1), 27. http://dx.doi.org/10.1186/s40537-021-00414-0.

dos Santos, A. A., & Gonçalves, W. N. (2019). Improving pantanal fish species recognition through taxonomic ranks in convolutional neural networks. *Ecological Informatics*, *53*, Article 100977. http://dx.doi.org/10.1016/j.ecoinf.2019.100977.

Schneider, S., Greenberg, S., Taylor, G. W., & Kremer, S. C. (2020). Three critical factors affecting automated image species recognition performance for camera traps. *Ecology and Evolution*, *10*(7), 3503–3517. http://dx.doi.org/10.1002/ece3.6147.

Shamsolmoali, P., Zareapoor, M., Shen, L., Sadka, A. H., & Yang, J. (2021). Imbalanced data learning by minority class augmentation using capsule adversarial networks. *Neurocomputing*, *459*, 481–493. http://dx.doi.org/10.1016/j.neucom.2020.01.119.

Shanmugam, V., Razavi-Far, R., & Hallaji, E. (2025). Addressing class imbalance in intrusion detection: A comprehensive evaluation of machine learning approaches. *Electronics*, *14*(1), 69. http://dx.doi.org/10.3390/electronics14010069.

Sharma, S., & Gosain, A. (2025). Addressing class imbalance in remote sensing using deep learning approaches: A systematic literature review. *Evolutionary Intelligence*, *18*(1), 23. http://dx.doi.org/10.1007/s12065-024-01012-8.

Sidumo, B., Sonono, E., & Takaidza, I. (2022). An approach to multi-class imbalanced problem in ecology using machine learning. *Ecological Informatics*, *71*, Article 101822. http://dx.doi.org/10.1016/j.ecoinf.2022.101822.

Simões, F., Bouveyron, C., & Precioso, F. (2023). DeepWILD: Wildlife identification, localisation and estimation on camera trap videos using deep learning. *Ecological Informatics*, *75*, Article 102095. http://dx.doi.org/10.1016/j.ecoinf.2023.102095.

Simović, P., Milosavljević, A., Stojanović, K., Radenković, M., Savić-Zdravković, D., Predić, B., Petrović, A., Božanić, M., & Milošević, D. (2024). Automated identification of aquatic insects: A case study using deep learning and computer vision techniques. *Science of the Total Environment*, *935*, Article 172877. http://dx.doi.org/10.1016/j.scitotenv.2024.172877.

Simović, P., Milosavljević, A., Stojanović, K., Savić-Zdravković, D., Petrović, A., Predić, B., & Milošević, D. (2025). The effects of data quality on deep learning performance for aquatic insect identification: Advances for biomonitoring studies. *Water*, *17*(1), 21. http://dx.doi.org/10.3390/w17010021.

Sohrab, F., & Raitoharju, J. (2020). Boosting rare benthic macroinvertebrates taxa identification with one-class classification. In *2020 IEEE symposium series on computational intelligence* (pp. 928–933). IEEE, http://dx.doi.org/10.1109/SSCI47803.2020.9308359.

Su, Z., Adam, A., & Nasrudin, M. F. (2025). Adaptive focal loss for keypoint-based deep learning detectors addressing class imbalance. *IEEE Access*, *13*, 31842–31856. http://dx.doi.org/10.1109/ACCESS.2025.3538917.

Sun, Y., Maeda, T. M., Solís-Lemus, C., Pimentel-Alarcón, D., & Buřivalová, Z. (2022). Classification of animal sounds in a hyperdiverse rainforest using convolutional neural networks with data augmentation. *Ecological Indicators*, *145*, Article 109621. http://dx.doi.org/10.1016/j.ecolind.2022.109621.

Sun, Y., Wong, A. K. C., & Kamel, M. S. (2009). Classification of imbalanced data: A review. *International Journal of Pattern Recognition and Artificial Intelligence*, *23*(04), 687–719. http://dx.doi.org/10.1142/S0218001409007326.

Tamouh, A., Tarik, M., Mniai, A., & Jebari, K. (2025). Fuzzy shadowed support vector machine for bankruptcy prediction. *Symmetry*, *17*(10), 1615. http://dx.doi.org/10.3390/sym17101615.

Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. In K. Chaudhuri, & R. Salakhutdinov (Eds.), *Proceedings of machine learning research*: *vol. 97*, *Proceedings of the 36th international conference on machine learning* (pp. 6105–6114). PMLR, URL: https://proceedings.mlr.press/v97/tan19a.html.

Teixeira, A. C., Ribeiro, J., Morais, R., Sousa, J. J., & Cunha, A. (2023). A systematic review on automatic insect detection using deep learning. *Agriculture*, *13*(3), 713. http://dx.doi.org/10.3390/agriculture13030713.

Thabtah, F., Hammoud, S., Kamalov, F., & Gonsalves, A. (2020). Data imbalance in classification: Experimental evaluation. *Information Sciences*, *513*, 429–441. http://dx.doi.org/10.1016/j.ins.2019.11.004.

Thölke, P., Mantilla-Ramos, Y.-J., Abdelhedi, H., Maschke, C., Dehgan, A., Harel, Y., Kemtur, A., Mekki Berrada, L., Sahraoui, M., Young, T., Bellemare Pépin, A., El Khantour, C., Landry, M., Pascarella, A., Hadid, V., Combrisson, E., O'Byrne, J., & Jerbi, K. (2023). Class imbalance should not throw you off balance: Choosing the right classifiers and performance metrics for brain decoding with imbalanced data. *NeuroImage*, *277*, Article 120253. http://dx.doi.org/10.1016/j.neuroimage.2023.120253.

Valan, M., Makonyi, K., Maki, A., Vondráček, D., & Ronquist, F. (2019). Automated taxonomic identification of insects with expert-level accuracy using effective feature transfer from convolutional networks. *Systematic Biology*, *68*(6), 876–895. http://dx.doi.org/10.1093/sysbio/syz014.

Venkataramanan, A., Laviale, M., Figus, C., Usseglio-Polatera, P., & Pradalier, C. (2021). Tackling inter-class similarity and intra-class variance for microscopic image-based classification. In *Lecture notes in computer science*: *vol. 12899*, *International conference on computer vision systems* (pp. 93–103). Springer, http://dx.doi.org/10.1007/978-3-030-87156-7_8.

Villon, S., Iovan, C., Mangeas, M., & Vigliola, L. (2022). Confronting deep-learning and biodiversity challenges for automatic video-monitoring of marine ecosystems. *Sensors*, *22*(2), 497. http://dx.doi.org/10.3390/s22020497.

Villon, S., Mouillot, D., Chaumont, M., Subsol, G., Claverie, T., & Villéger, S. (2020). A new method to control error rates in automated species identification with deep learning algorithms. *Scientific Reports*, *10*, 10972. http://dx.doi.org/10.1038/s41598-020-67573-7.

Wahab, N., Khan, A., & Lee, Y. S. (2017). Two-phase deep convolutional neural network for reducing class skewness in histopathological images based breast cancer detection. *Computers in Biology and Medicine*, *85*, 86–97. http://dx.doi.org/10.1016/j.compbiomed.2017.04.012.

Walsh, R., & Tardy, M. (2023). A comparison of techniques for class imbalance in deep learning classification of breast cancer. *Diagnostics*, *13*(1), 67. http://dx.doi.org/10.3390/diagnostics13010067.

Wang, L., Chen, W., Yang, W., Bi, F., & Yu, F. R. (2020). A state-of-the-art review on image synthesis with generative adversarial networks. *IEEE Access*, *8*, 63514–63537. http://dx.doi.org/10.1109/ACCESS.2020.2982224.

Wang, J., Lukasiewicz, T., Hu, X., Cai, J., & Xu, Z. (2021). RSG: A simple but effective module for learning imbalanced datasets. In *2021 IEEE/CVF conference on computer vision and pattern recognition* (pp. 3783–3792). http://dx.doi.org/10.1109/CVPR46437.2021.00378.

Wang, J., & Perez, L. (2017). The effectiveness of data augmentation in image classification using deep learning. *Convolutional Neural Networks Vis. Recognit*, *11*(2017), 1–8.

Wongvorachan, T., He, S., & Bulut, O. (2023). A comparison of undersampling, oversampling, and SMOTE methods for dealing with imbalanced classification in educational data mining. *Information*, *14*(1), 54. http://dx.doi.org/10.3390/info14010054.

Wu, Q., Chen, Y., & Meng, J. (2020). DCGAN-based data augmentation for tomato leaf disease identification. *IEEE Access*, *8*, 98716–98728. http://dx.doi.org/10.1109/ACCESS.2020.2997001.

Xu, C., Cai, R., Xie, Y., Cai, H., Wang, M., Gao, Y., & Ma, X. (2022). Fine-grained butterfly recognition via peer learning network with distribution-aware penalty mechanism. *Animals*, *12*(20), 2884. http://dx.doi.org/10.3390/ani12202884.

Yang, Q., Feng, X., & Tan, S. (2025). One-stage long-tailed object detection based on adaptive class suppression equalized focal loss. *IEEJ Transactions on Electrical and Electronic Engineering*, *20*, 1424–1435. http://dx.doi.org/10.1002/tee.70027.

Yang, L., Jiang, H., Song, Q., & Guo, J. (2022). A survey on long-tailed visual recognition. *International Journal of Computer Vision*, *130*(7), 1837–1872. http://dx.doi.org/10.1007/s11263-022-01622-8.

Yang, J., & Li, J. (2017). Application of deep convolution neural network. In *2017 14th international computer conference on wavelet active media technology and information processing* (pp. 229–232). IEEE, http://dx.doi.org/10.1109/ICCWAMTIP.2017.8301485.

Yang, D.-Q., Li, T., Liu, M.-T., Li, X.-W., & Chen, B.-H. (2021). A systematic study of the class imbalance problem: Automatically identifying empty camera trap images using convolutional neural networks. *Ecological Informatics*, *64*, Article 101350. http://dx.doi.org/10.1016/j.ecoinf.2021.101350.

Yu, W., Yang, K., Bai, Y., Xiao, T., Yao, H., & Rui, Y. (2016). Visualizing and comparing AlexNet and VGG using deconvolutional layers. In *JMLR workshop and conference proceedings*: *vol. 48*, *Proceedings of the 33rd international conference on machine learning (ICML)*. New York, NY, USA.

Yue, G., Wei, P., Liu, Y., Luo, Y., Du, J., & Wang, T. (2023). Automated endoscopic image classification via deep neural network with class imbalance loss. *IEEE Transactions on Instrumentation and Measurement*, *72*, 1–11. http://dx.doi.org/10.1109/TIM.2023.3264047.

Zaghloul, A., Saber, M., Gadow, S., & Awad, F. (2020). Biological indicators for pollution detection in terrestrial and aquatic ecosystems. *Bulletin of the National Research Centre*, *44*(1), 1–11. http://dx.doi.org/10.1186/s42269-020-00385-x.

Zhang, Y., Kang, B., Hooi, B., Yan, S., & Feng, J. (2023). Deep long-tailed learning: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *45*(9), 10795–10816. http://dx.doi.org/10.1109/TPAMI.2023.3268118.

Zhao, Z., Alzubaidi, L., Zhang, J., Duan, Y., & Gu, Y. (2024). A comparison review of transfer learning and self-supervised learning: Definitions, applications, advantages and limitations. *Expert Systems with Applications*, *242*, Article 122807. http://dx.doi.org/10.1016/j.eswa.2023.122807.

Zhou, Z.-H. (2021). Ensemble learning. In *Machine learning*. Singapore: Springer, http://dx.doi.org/10.1007/978-981-15-1967-3_8.

Zhou, B., Cui, Q., Wei, X.-S., & Chen, Z.-M. (2020). BBN: Bilateral-branch network with cumulative learning for long-tailed visual recognition. In *2020 IEEE/CVF conference on computer vision and pattern recognition* (pp. 9716–9725). http://dx.doi.org/10.1109/CVPR42600.2020.00974.

Zhuang, F., Qi, Z., Duan, K., Xi, D., Zhu, Y., Zhu, H., Xiong, H., & He, Q. (2021). A comprehensive survey on transfer learning. *Proceedings of the IEEE*, *109*(1), 43–76. http://dx.doi.org/10.1109/JPROC.2020.3004555.